Do Language Models Have Educational Personalities? Behavioral Differences, Stability, and Prompt Control

Anonymous ACL submission

Abstract

Understanding AI tutors requires knowing both whether they answer correctly and how they usually help students. We study *educational personality*: recurring, observable teaching tendencies. Our investigation begins with selected subsets of over one million educational evaluation records and continues with targeted experiments and tests on new responses. Models show recurring differences in how directly they help and how much information they present, with stronger evidence for directness. Other measures show shared responses or differences limited to particular settings. Knowing how often a model previously gave answers or invited student reasoning helps predict these actions on new problems, including under tested changes in student wording. For acknowledgement of feeling or difficulty, knowing how each model responds to student cues improves prediction beyond its usual rate, but this extra benefit is lost under new wording. Explicit teaching instructions make models perform the same required action while leaving other choices different. These findings help educators understand and compare AI tutors through their teaching habits, when those habits remain predictable, and how they change with instructions. The findings also help developers investigate model selection, prompt design, and additional system support.

1 Introduction

An AI tutor can answer correctly while choosing different ways to help: explain the solution, invite the student to attempt a step, or combine both. For an educator selecting a tutor, correctness therefore leaves a practical question open: how will this model usually respond, and when can that behavior be expected to recur?

Our investigation draws on 1,028,822 raw evaluation records across 39 educational benchmark settings, including tutoring, assessment, and problem solving. Looking at models answering the same educational questions raised an initial question: do their apparently different teaching styles recur across responses? We investigate this using subsets suited to each analysis. We use *educational personality* to investigate observable teaching tendencies, without assuming human personality traits. Initial screening found that some differences recur while others vary. We therefore ask what happens when the problems, student wording, or teaching instructions change.

We ask three questions: **RQ1: Which educational behaviors exhibit recurring model differences? RQ2: Under which input changes do those differences persist? RQ3: How do explicit instructions reshape the differences?**

Some teaching tendencies distinguish models more consistently than others. Across tasks, models differ in how directly they help and how much information they present. Exploratory archive analyses also find recurring differences in how confidently they state answers. Other measures show shared responses or insufficient evidence of consistent differences.

Knowing how often each model gave answers or invited reasoning helps predict these actions on new problems, even when students use different wording. For acknowledgement of feeling or difficulty, knowing how a model responds to student cues further improves prediction, but this extra benefit is lost under new wording. Giving models the same teaching instruction makes required actions converge while other choices remain different. These remaining differences need not preserve the original model ranking or action rates.

Prior work connects personality consistency, prompt sensitivity, and tutor action mixtures (Lee et al., 2025; Kucheria et al., 2025; Liu et al., 2026; Lee et al., 2026). We connect these descriptions of teaching habits to tests of when they remain predictable. We fix predictions before collecting new responses, then test whether previous behavior helps predict answers to new inputs. We also compare models on the same inputs under different teaching instructions to see how deliberate control changes their behavior. Section 3 identifies which analyses are exploratory and which predictions were fixed in advance.

For educators, these findings describe how a tutor usually helps, when that behavior can be anticipated, and how it responds to instructions. A recurring habit is not necessarily the best choice for every activity. Developers can use such evidence to investigate model selection and additional system support. These applications follow from the behavioral findings; learning benefits and harness effectiveness remain untested.

2 Related Work

Personality and situated behavior. TRAIT extends personality inventories into scenario-based choices and evaluates distinctiveness, consistency, and prompting effects, including agreement between original and paraphrased items (Lee et al., 2025). Behavioral Mode Axes examines choices across interaction registers, evaluates open-ended generations, and studies activation-based control (Liu et al., 2026). Internal-feature interventions and conversational studies further connect personality-related behavior to changes in situations (Zhang et al., 2026; Han et al., 2026). PERSONA composes activation vectors for context-dependent personality control (Feng et al., 2026), while Dynamic Persona Coherence distinguishes stable identity from appropriate state changes in role-playing (Qi et al., 2026). These studies connect consistency, context, and control. We examine that relationship in existing AI tutors by predicting observable teaching actions before collecting new responses and testing how shared instructions change them.

Tutor behavior and educational personas. The CIMA comparison reports characteristic action mixtures and response complexity for three LLMs in language-learning dialogues; its generation prompt requests both a response and action labels (Kucheria et al., 2025). Tutor-persona steering learns from human tutoring dialogues and evaluates persona alignment on held-out dialogues (Lee et al., 2026). These studies establish tutor-specific styles and methods for learning them. We test whether models' past actions help predict new responses, whether the same forecasts work when students change their wording, and how shared instructions change both required actions and other choices. We code visible answers after generation under controlled student cues and teaching instructions.

Teaching quality and behavioral predictability. MRBench and MathTutorBench evaluate pedagogical capabilities (Maurya et al., 2025; Macina et al., 2025), while PATS studies teaching adapted to student personality (Rooein et al., 2026). We complement capability evaluation by asking which teaching actions can be predicted from a model's previous responses and when changes in input make those predictions less useful. This connects descriptions of teaching habits with evidence about what tutors are likely to do next.

3 Study Design and Tests

A *behavioral profile* estimates how often a model performs an action. A default profile describes its usual rate; a conditional profile also accounts for how it responds to student cues. We examine rated teaching behaviors, expressed confidence, safety boundaries, affiliation, and specific actions. These analyses use different model groups and measures (Appendix A); their samples overlap and cannot be added as independent observations.

What stability means. We ask whether a model repeats an action on the same input, whether models show similar relative tendencies across tasks, and whether past behavior helps predict new responses. These are separate questions. Showing that action rates stay unchanged requires an equivalence test. Likewise, agreement among coders and consistency of model differences across tasks use different ICC measures.

3.1 Exploration and Prospective Design

Analyses of selected archive subsets identify candidate teaching tendencies. The archive spans educational roles beyond tutoring, and historical prompt and deployment provenance is incomplete (Appendix B). An excluded 160-answer pilot develops the prospective event measures (Appendix C).

The new-response experiments test MiniMax-M3, MiniMax-M2.7, GLM-5.3, DeepSeek-V4-Pro, and Doubao-Seed-2.0-Lite deployment configurations. Earlier records for GLM-5.2 and Doubao-2.0-Pro are analyzed separately. There are 32 training questions in 30 conservative source families and 32 confirmation questions in 32 separate families, with eight questions per partition in mathematics, science, language, and logic/data reasoning.

Research agents authored the English problems and student messages; two model reviewers checked all 64 packages before freezing. Related training questions stay grouped, and pilot templates and numerical substitutions are excluded. The holdout tests new sources within these domains, not unseen domains.

For each question, the student shows either an erroneous attempt or correct unfinished work, and states feeling either calm or frustrated. The two work states differ in both correctness and progress. Every main condition gives the tutor the same reference solution, marked as unseen by the student. We request each response twice and group statistical comparisons by source problem; repetitions do not add independent sources. Appendix D records materials, deployment identities, settings, and counts.

3.2 Observable Actions and Recurrence (RQ1)

We focus on three actions that can be identified in the text and showed agreement among coders on their presence: *answer revelation*, an explicit final resolution regardless of correctness; *reasoning elicitation*, an invitation for student reasoning not immediately answered by the tutor; and *affect acknowledgement*, explicit recognition of feeling or difficulty beyond generic praise or politeness. A reply can contain several of these actions; we do not treat them as independent personality factors. Five secondary events are retained regardless of results.

Three models—MiniMax-M3, DeepSeek-V4-Pro, and GLM-5.3—label the actions in each response using its context, reference, and original answer lines. Generator identity and design labels are withheld, although style may reveal identity. Each positive label must cite nonempty evidence lines. The majority of the three coders determines the action label. Repairs address incomplete or malformed outputs, never disagreement or label choices. Evidence lines and coder agreement alone do not guarantee correct labels. RQ1 reports action rates and agreement between repeated requests, separating agreement on an action being present from agreement on its absence. Statistical comparisons account for replies sharing a source problem.

3.3 Predicting Responses to New Problems (RQ2)

We use logistic regression, not another language model, to predict the actions of the same five deployments on new questions. We fit it to coded training answers using model identity, subject, and scripted student state, and fix all forecasts before collecting confirmation responses.

We make two comparisons. First, does knowing the model's usual behavior improve prediction over knowing only the subject and student state? Second, does knowing how each model responds to student cues improve prediction over its usual behavior within a subject? We call the reductions in Brier loss the *default gain* and *state gain*, respectively; positive gains mean lower prediction error. The six first-stage tests use Holm correction. Appendix D.4 gives fitting and inference details.

3.4 Transfer Across Student Expressions (RQ2)

To test whether wording changes affect prediction, an extension reuses the 32 confirmation problems, original forecasts, and five deployments. We fix this protocol after seeing the first confirmation results but before collecting the extension responses. Original wording generated again, new explicit synonyms, implicit cues, and a peer-context control each yield 1,280 answers: 5,120 additional replies, for 10,440 across the two stages, excluding the pilot. The primary tests use the original fitted predictors. This extension tests new expressions on the same problems, not another independent set of problems.

The explicit and implicit conditions each use four fixed cue pairs. Each pair is assigned to two sources per domain; not every phrase is tested on every problem. Implicit cues also change actions, willingness, or time information; their effects cannot be attributed to emotion alone. The peer arm keeps the current student calm while quoting a calm or frustrated classmate working on another problem.

Six primary tests cover default revelation and elicitation and conditional acknowledgement in the two new-expression arms. Paired source-level tests use Holm correction within this family and 10,000 domain-stratified source bootstrap draws. Repeated original wording, contrasts against it, and peer effects are secondary. Appendix G reports exact cues, coder and generator-deletion sensitivities, and measurement recovery.

3.5 Instruction Control and Sensitivities (RQ3)

On 16 hash-selected sources, each model receives every student state under the canonical role, two neutral paraphrases, ASK, and EXPLAIN. Paired comparisons measure how much action rates change and whether the changes differ by model or student state. Using an equivalence margin of ± 0.10, our source-level check is too imprecise to establish equivalent effects of neutral paraphrases beyond these sixteen sources. Nonsignificance does not establish stability. Joint revelation/elicitation rates allow co-occurrence without inferring order; instruction effects do not measure better teaching.

Checks specified before confirmation use eight sets of coder labels, exclude each of the five generating models in turn, and resample training sources 200 times. Content-error screens remove complete five-model comparison slots without refitting forecasts. Low reviewer agreement limits their interpretation. Full specifications, diagnostic controls, coverage rules, and amendments appear in Appendix D.

A retrospective MathTutorBench extension (Macina et al., 2025) compares general scaffolding and explicit probing, both limited to two sentences, on 256 exact source groups from seven historical deployments. Five-fold source-held-out Jeffreys rates test same-instruction and other-instruction model profiles against an instruction-only baseline. Two provider-blocked codes remain missing; agreeing available votes identify every affected majority. The post-hoc amendment preserves missing labels and adds analyses on 254 fully coded sources (Appendix K).

4 Which Teaching Tendencies Distinguish Models?

Some teaching behaviors show recurring differences between models; others show differences only in certain settings, shared responses, or inconclusive results (RQ1; Table 1). We report the evidence for each behavior separately.

4.1 Helping Choices Show Recurring Model Differences

Models differ in how directly they help and how often they invite student reasoning. On the 32 source problems under the common tutoring instruction, GLM reveals an answer in 22.3% of responses and invites reasoning in 96.5%; DeepSeek and Doubao reveal answers in every response, with reasoning invitations below 5%. M3 falls between these models on both actions (Appendix Table 10). These rates describe usual behavior under the same role and inputs; they do not rank teaching quality or assign models to fixed types. Giving an answer and inviting reasoning are separately coded actions: an invitation leaves a content-related reasoning step for the student rather than asking a question immediately answered by the tutor. A reply may contain both actions without establishing their order or a multi-turn teaching process.

Ratings of assistance directness provide complementary evidence. The scale runs from hints or questions to complete solutions, and model differences recur across tasks (ICC 0.629). To pass the strongest screening rule, a dimension must be measured reliably, distinguish models across tasks, and receive further support from teaching actions, instruction changes, or diagnosis outcomes. Only directness passes the full rule. Its changes under instructions agree with independently inferred teaching actions, and the direction remains the same when each coder is excluded (Appendix A). The directness ratings and binary action labels describe related but different aspects of helping. Section 5 tests whether the action labels support prediction on new inputs.

4.2 Other Differences Have More Limited Support

Models also differ across tasks in the amount of information and number of steps they present (ICC 0.663). These response-complexity ratings pass the cross-task consistency criteria, but lack the additional evidence needed to pass the full screening rule. They measure what a response contains, not the mental effort a student experiences.

Differences in expressed confidence recur across four context families in six historical configurations (ICC 0.852). This exploratory archive result concerns how confidently models report their answers; it does not establish correctness or calibration. Revision and abstention measures show mixed consistency, so expressed confidence alone does not establish a broader cautiousness trait. Safety measures also show recurring differences within some tasks, but do not support one ranking of how permissive models are across tasks. These results apply to the reported measures and settings; they have not received the same new-response tests as the three

Finding type	Representative evidence	Interpretation
Recurring helping differences	Answer revelation and reasoning invitations; complementary directness screen	Some helping tendencies distinguish models under the tested conditions
More limited recurrence	Response complexity, expressed confidence, and within-task safety patterns	Recurrence has different scopes across behaviors and measures
Shared responses	Identical default affiliation-inventory choices	These fixed-choice items do not distinguish the tested models
Unresolved consistency	Mixed revision/abstention results; failed candidate checks	Insufficient support for recurrence does not establish absence of differences

Table 1: Types of findings in Section 4. Shared responses and unresolved consistency are distinguished within the third subsection. The full evidence map, including subsequent persistence tests, is in Appendix Table 2.

action measures (Appendix A).

4.3 Shared Responses and Uncertain Differences

Some measures yield common responses. In the affiliation pilot, all five models choose the affiliative option on every default questionnaire item. Those choices do not distinguish models. Their freely written responses do show recurring differences within this pilot, but affiliation ratings closely overlap with ratings of surface warmth. The shared questionnaire choices therefore do not mean that the models behave identically in open responses, and the pilot does not establish a separate general affiliation trait (Appendix A).

Other measures fall short for different reasons. Ratings of how actionable the next step is change with instructions but fail the cross-task stability check. The rated elicitation scale also fails its full screening rule. The later predictive success of a separate binary measure— whether a reply invites reasoning—does not validate that scale. We retain these failed checks without treating them as proof that models never differ. The next experiments focus on predicting revelation, elicitation, and acknowledgement and testing how instructions change them. They do not validate every tendency examined in the broader screen.

5 Which Differences Persist When Inputs Change?

5.1 A Teaching Habit Need Not Appear in Every Reply

Past behavior can help predict a model's next response even when it does not repeat the same action every time. M3 agrees with itself on whether to invite reasoning in only 57.8% of 128 matched input pairs (95% source-bootstrap interval 50.0–66.4%). High agreement can also reflect repeated absence: DeepSeek and Doubao have elicitation agreement above 90%, but none of their few invitations recur in both requests. They agree perfectly on answer revelation because they always reveal an answer in this sample.

Agreement therefore needs to be read alongside how often each action occurs and whether earlier responses help predict later ones (Appendix F). Appendix E illustrates the codes on matched replies.

5.2 Past Behavior Helps Predict New Responses (RQ2)

Knowing the model's usual behavior improves prediction of all three actions on new problems, compared with knowing only the subject and student state. Brier loss decreases by 0.091 for answer revelation, 0.130 for reasoning elicitation, and 0.020 for acknowledgement; all three gains pass the six-test Holm correction. Thus, models' previous teaching choices provide information that the scripted educational context alone does not supply.

Knowing how each model responds to student cues further improves acknowledgement predictions over its usual rate within a subject. The same comparison gives no clear additional benefit for revelation or elicitation. Full estimates and uncertainty intervals appear in Appendix Tables 11 and 12.

A predictor using training-response length and formatting makes larger errors than the model-default predictor for all three actions (Appendix Table 11). These two style measures do not match the predictive performance of model defaults. Richer style measures may do better, and the comparison does not identify a causal effect of model identity.

5.3 Prediction Under New Student Wording (RQ2)

When students express themselves differently, the original model defaults still help predict whether tutors give answers or invite reasoning. For acknowledgement, the extra benefit of knowing each model's response to student cues is lost. This benefit remains positive when we collect new replies using the original wording during the same period. The contrast weakens an explanation based only on a general loss of predictive value over time. Figure 1 shows the estimates and uncertainty; Appendix Table 16 gives all six tests and Appendix G the paired contrasts.

The acknowledgement gains have the same sign pattern when each coder's labels are used separately and when each generating model is left out, although some intervals include zero. These checks reuse the same responses. The result concerns prediction: models still acknowledge feeling or difficulty at different rates for contrasting student cues. It does not mean that models stop responding to student cues. The implicit condition also changes more than affect wording.

Alternative explanations. Losing an advantage over a baseline does not always mean making larger errors. Under explicit synonyms, the cue-based predictor's own loss improves slightly, yet it no longer outperforms the baseline. Under implicit cues, its own loss also worsens. Further checks show that a common change in acknowledgement frequency cannot explain the contrast, and the tested logistic recalibrations do not restore a positive gain. These post-result checks narrow the explanations without identifying a unique mechanism (Appendices I–J).

In the secondary peer control, changing a classmate's quoted feelings changes acknowledgement rates even though the current student remains calm. Selected examples show this can happen when the model correctly distinguishes the two students' feelings. The contrast alone therefore does not establish confusion about whose feelings are being expressed (Appendix G).

5.4 Useful Predictions Do Not Require Unchanged Action Rates

A model's action rates can change while its earlier behavior still helps predict new responses. Neutral role paraphrases produce some observed shifts, and our sample is too limited to establish equivalence (Appendix F). The transfer tests show that previous behavior remains informative under the tested changes. They do not show that rates or model rankings stay fixed. Default and state gains also compare different predictors, so their sizes cannot be read as a common stability score. Persistence across provider updates remains untested.

5.5 Checking Dependence on Coders, Models, and Answer Errors

The main pattern holds across eight sets of coder labels: model defaults improve prediction, and cue-specific responses add value mainly for acknowledgement. Resampling training sources and checking missing codes preserve this pattern. These checks reduce concern that the findings depend on one coding choice, but do not establish that the labels are correct (Appendix F).

The strength of the differences depends on which models are compared. Removing GLM substantially reduces default gains for revelation and elicitation. Removing M2.7 leaves a small default acknowledgement gain whose interval spans zero. The full-panel result therefore need not hold equally strongly for every subset of models.

Default gains and the acknowledgement state gain remain positive after removing matched response groups flagged for content errors. However, reviewers often disagree about these errors. The screen cannot guarantee correct retained answers or rule out differences in task ability (Appendix F.5).

6 How Do Instructions Reshape the Differences?

When models follow the same instruction, do they behave alike throughout the response? RQ3 examines which actions become similar, which remain different, and whether past behavior still helps predict responses under a new instruction.

6.1 Models Agree on One Action but Differ on Others

On the same 16 problems, ASK requires a next step without the final answer, whereas EXPLAIN requires a solution and allows follow-up reasoning. All five deployments invite reasoning in every ASK response and reveal the answer in every EXPLAIN response (Figure 2). Yet asking for reasoning does not guarantee that the answer is withheld: under ASK, M2.7 and M3 still reveal answers in 9/128 and 4/128 replies, respectively; the other three models never do so in this sample.

Under EXPLAIN, the models differ in whether they also invite student reasoning. GLM and M3 invite reasoning in 62.5% and 45.3% of their explanations, compared with 1.6–5.5% for the others. Thus, all models perform the required action, but they differ in what else they include in the response. These codes identify reasoning invitations, not their

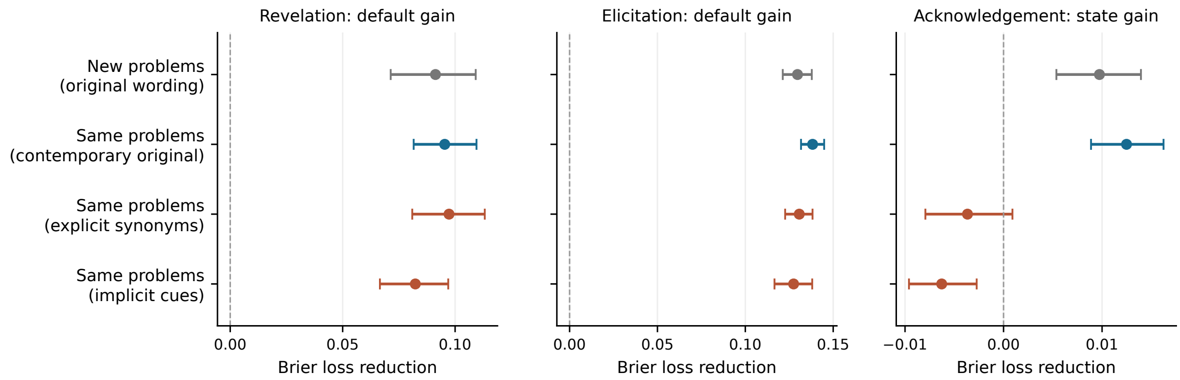

Figure 1: Transfer depends on what is changed. First-stage confirmation tests new source problems with original wording. The three extension rows reuse those same 32 problems and original forecasts. Default revelation and elicitation gains persist, while the conditional acknowledgement gain loses its positive sign under new expressions. Bars are source bootstrap 95% intervals conditional on frozen fits, not four independent replications or simultaneous intervals. Horizontal scales differ by panel.

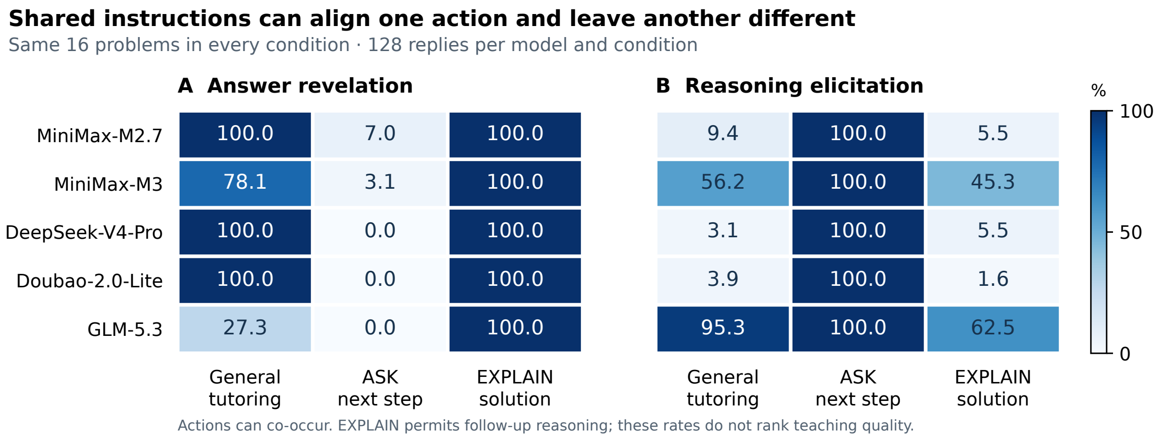

Figure 2: One shared instruction can align one action while leaving another different. Each cell is a descriptive percentage of 128 replies on the same 16 problems; this matched subset differs from the 32-problem default table. ASK requires a next step without the final answer. EXPLAIN requires the answer and permits follow-up reasoning. All models reveal under EXPLAIN, yet elicitation ranges from 1.6% to 62.5%. Actions may co-occur; these frequencies are not quality scores. Full prompt comparisons and uncertainty limitations appear in Appendix F.

order or teaching quality; full comparisons appear in Appendix F.

6.2 Instructions Can Change Model Rankings

In the historical ratings, a teaching instruction can change which model is more direct. Across both standard and hard MathDial benchmark variants (Macina et al., 2025), Doubao-Seed-2.0-Pro is more direct than GLM-5.2 under the generic instruction, but less direct under the same pedagogy instruction. This ordering changes in the observed means; we did not separately establish the pairwise reversals with significance tests.

This six-model analysis was specified after inspecting results. It also finds that instructions change average directness, elicitation ratings, and response complexity by amounts comparable to the original differences between models. Models respond differently to the instruction on directness and elicitation in both variants, even after accounting for how far their ratings could move before reaching a scale boundary. Complexity does not pass this stricter check. Together, these results motivate comparing models under the intended teaching instruction as well as at baseline. Detailed estimates and supplementary model-attribution analyses remain in Appendix A.4.

6.3 Predictions May Fail Under a Different Instruction

Past responses can help predict behavior under the same instruction but make prediction worse under a different one. In the paired archive of seven historical configurations, knowing the model improves prediction of revelation and elicitation under general scaffolding. Under explicit probing, nearly every reply invites reasoning, so knowing the model adds less. Using a model's rates from the other instruction makes prediction worse than using only the current instruction. This holds in both directions (Appendix K).

Thus, a model's past behavior can remain useful when student wording changes (Section 5) yet fail to predict responses under a different teaching instruction. Evaluating a tutor requires knowing both its usual habits and how those habits change with instructions.

7 Discussion and Educational Implications

Teaching habits help educators anticipate responses. Knowing a model's past teaching choices helps predict its responses to new problems. For giving answers and inviting reasoning, this remains useful under the tested changes in student wording. Educators can therefore form expectations from patterns across responses, rather than isolated examples. The support varies by behavior: complexity and confidence recur in more limited settings, shared affiliation-questionnaire choices do not distinguish models, and a predictable habit need not appear in every reply.

Teaching habits complement capability evaluation (Maurya et al., 2025; Macina et al., 2025). Knowing that a model can solve a problem does not tell us whether it usually explains the answer or asks the student to attempt a step. Neither choice suits every activity. A behavioral profile helps educators compare how a model usually responds with what an activity requires; it does not rank tutors by their effects on learning.

Following an instruction does not settle every teaching choice. Under EXPLAIN, every model gives the answer, but some also invite reasoning much more often than others. The same required action need not produce the same overall response. These remaining differences do not prove that models preserve their original styles: their rankings can change, and rates learned under one instruction may not predict another.

Reliability therefore needs to cover the full response under the intended instruction. Under ASK, some replies invite a step but also give away the answer. Checking only for the invitation would miss a failure that matters when students are meant to attempt the step themselves. Evaluation should check all behaviors required by the activity and the input changes likely to occur. A predictable habit or success on one requirement is not enough.

Harness design should start from the model's behavior. Developers can identify what needs additional control by testing a model's usual habits, their persistence under changed inputs, and its response to instructions. A workflow or checking mechanism—a *harness*—may be reusable across models, but its settings and effectiveness need to be tested with each model it supports. Evaluation can show which desired behaviors a simple prompt already produces reliably and which important failures remain.

Developers can then test whether the configured system closes those gaps. When the model changes, they need to check these behaviors again. Harness benefits and learning gains remain untested; the findings help specify what additional support should address and how to evaluate it.

8 Conclusion

Language models show recurring teaching tendencies, but the evidence differs across behaviors. Helping choices have the strongest combined support; other measures show differences within narrower settings, shared responses, or mixed results. Knowing how often a model gives answers or invites reasoning helps predict these actions on new problems, even under tested changes in student wording. For acknowledgement, knowing how each model responds to student cues adds predictive value beyond its usual rate on new problems, but this extra benefit is lost under new wording. Explicit instructions can make models perform the same required action while leaving other choices different.

These findings help educators compare tutors through their teaching habits alongside task capability. They also help developers investigate which gaps between a model's behavior and an activity's requirements need better prompts or additional system support.

Limitations

Coverage of settings and deployments. The prospective tests examine five selected deployments on agent-authored English problems with scripted student messages. The 32 confirmation source groups support controlled comparisons, but cover a limited range of educational situations. Repeated replies do not broaden that coverage. Extending these findings to real classrooms, multilingual teaching, longer interactions, and additional models requires further study. Short-term repetition also leaves open whether the observed tendencies persist across provider updates.

Automated behavioral measurement. The main predictive findings hold across the tested sets of coder labels, but automated coders may share biases that our agreement and sensitivity checks cannot fully detect. Independent human validation would strengthen the interpretation of the behavioral measures. Reviewers also disagree about errors in the generated answers. This limits what the error screen can tell us about whether teaching tendencies reflect differences in task ability (Appendix F.5). It does not directly measure the accuracy of the primary behavioral codes.

Coverage of teaching tendencies. The tested measures do not constitute a comprehensive educational personality profile. We have not validated measures of adaptive feedback, pacing over multiple turns, or relationships among tendencies. The evidence therefore supports selected teaching tendencies within their reported conditions, not a complete account of a model's teaching behavior.

Ethical Considerations

The prospective student messages and problems are agent-authored stimuli, not interactions with recruited learners. Behavioral profiles describe model outputs; they should not be used to diagnose students' emotions or infer human psychological traits in a model. Favorable-looking teaching habits do not establish correctness, learning benefit, or suitability for every learner. Applying profiles beyond the tested language, tasks, and deployments could mislead educators or overlook needs absent from the evaluation. Educational use requires checking the intended activity and population; our harness implications remain proposals for evaluation.

Acknowledgements

We used Codex and Claude Code as AI assistants. Codex supported language polishing and manuscript revision. AI assistance also covered research planning, stimulus construction, programming, analysis, literature search, and development of manuscript text from research discussions. Model-based behavioral coding and content review are documented in Section 3 and the appendices. The human authors remain responsible for the methods, results, citations, and text.

References

Xiachong Feng, Liang Zhao, Weihong Zhong, Yichong Huang, Yuxuan Gu, Lingpeng Kong, Xiaocheng Feng, and Bing Qin. 2026. PERSONA: Dynamic and compositional inference-time personality control via activation vector algebra. In *The Fourteenth International Conference on Learning Representations*.

Bin Han, Deuksin Kwon, and Jonathan Gratch. 2026. Personality expression across contexts: Linguistic and behavioral variation in LLM agents. *Preprint*, arXiv:2602.01063.

Wassily Hoeffding. 1963. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30.

Aayush Kucheria, Nitin Sawhney, and Arto Hellas. 2025. Comparing behavioral patterns of LLM and human tutors: A population-level analysis with the CIMA dataset. In *Proceedings of the 20th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2025)*, pages 873–881.

Jaewook Lee, Alexander Scarlatos, Simon Woodhead, and Andrew Lan. 2026. Letting tutor personas speak up for LLMs: Learning steering vectors from dialogue via preference optimization. In *Proceedings of the 21st Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2026)*, pages 78–92.

Seungbeen Lee, Seungwon Lim, Seungju Han, Giyeong Oh, Hyungjoo Chae, Jiwan Chung, Minju Kim, Beong-woo Kwak, Yeonsoo Lee, Dongha Lee, Jinyoung Yeo, and Youngjae Yu. 2025. Do LLMs have distinct and consistent personality? TRAIT: Personality testset designed for LLMs with psychometrics. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 8412–8452.

Ning Li, Zheng Zhang, Zhenya Huang, Rui Li, Yi Zhan, Yinbo Luo, Qi Liu, and Enhong Chen. 2026. LongTutor: Benchmarking large language models for long-term personalized tutoring. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 29712–29737.

Haoze Liu, Run Liu, Haiying Xu, Jiahui Han, Siyuan Fang, Siyu Yan, Huiqi Deng, Guanchu Wang, and Na Zou. 2026. Your LLM, your style: Behavioral mode axes for LLM behavioral control. *Preprint*, arXiv:2608.10703.

Jakub Macina, Nico Daheim, Sankalan Chowdhury, Tanmay Sinha, Manu Kapur, Iryna Gurevych, and Mrinmaya Sachan. 2023. MathDial: A dialogue tutoring dataset with rich pedagogical properties grounded in math reasoning problems. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 5602–5621.

Jakub Macina, Nico Daheim, Ido Hakimi, Manu Kapur, Iryna Gurevych, and Mrinmaya Sachan. 2025. MathTutorBench: A benchmark for measuring open-ended pedagogical capabilities of LLM tutors. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 204–221.

Kaushal Kumar Maurya, Kv Aditya Srivatsa, Kseniia Petukhova, and Ekaterina Kochmar. 2025. Unifying AI tutor evaluation: An evaluation taxonomy for pedagogical ability assessment of LLM-powered AI tutors. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 1234–1251.

Yirui Qi, Xiaoming Zhang, Ruilin Zeng, Mengyao Liu, Ziyi Zhou, Dezhuang Miao, Bingyu Yan, and Zhenyu Guan. 2026. Beyond static persona consistency: Dynamic persona coherence in LLM role-playing. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 28942–28956.

Donya Rooein, Sankalan Pal Chowdhury, Mariia Eremeeva, Yuan Qin, Debora Nozza, Mrinmaya Sachan, and Dirk Hovy. 2026. PATS: Personality-aware teaching strategies with large language model tutors. In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 4186–4211.

Rose Wang, Qingyang Zhang, Carly Robinson, Susanna Loeb, and Dorottya Demszky. 2024. Bridging the novice-expert gap via models of decision-making: A case study on remediating math mistakes. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2174–2199.

Ruikang Zhang, Shuo Wang, and Qi Su. 2026. From representations to behaviors: Exploring the person–situation–behavior triad in LLMs. *Preprint*, arXiv:2607.26853.

A Evidence for the Broader Behavioral Findings

The following analyses and experiments belong to the present investigation. They retain different sampling frames, deployment aliases, and decision rules. Table 2 gives the full evidence map, including the persistence results developed after Section 4. Table 3 supplies quantitative anchors for the broader behavioral findings. Its accompanying inventory maps each quantitative row to a saved aggregate, field, and source hash. These tables report existing results; they do not introduce a new analysis or a common validation rule.

A.1 Semantic Screening and Candidate Dimensions

The semantic panel rates visible responses from six historical configurations on eight dimensions using three model coders. Assistance directness spans hint/question-only assistance to a complete solution; cognitive load spans a manageable next step to many simultaneous steps or concepts. Thus the latter is an output rubric, not a learner measurement. Ratings are examined for coder agreement, variation, cross-task consistency, and links to observable actions and prompt changes. Coders' response-level agreement and models' cross-task consistency are separate quantities. Directness alone satisfies the complete saved behavioral-disposition rule. Cognitive load satisfies the narrower semantic-signature criteria; action/quality evidence does not support its promotion to a fully validated disposition. No learning-efficacy claim follows from the directness decision either.

The decision rule was fixed after partial coder results had been inspected, before the remaining coder reveal; it is not a fully outcome-blind preregistration. It constrains subsequent decisions without converting the whole semantic screen into independent prospective confirmation.

Complete screening rule. The hierarchy requires all preceding tiers; passing one component alone does not validate a dimension. Reliable measurement requires coder ICC(3,k) $\geq$ .60, consensus-score SD $\geq$ .50, neither endpoint above 80%, minimum leave-one-coder model-profile Spearman $\geq$ .70, and no leave-one-coder prompt-effect sign reversal among headline model–task contrasts. Cross-task signature status additionally requires a positive lower 95% bootstrap bound for model variance in every sampled benchmark and either cross-task ICC(3,1) $\geq$ .50 or median pairwise-task Spearman $\geq$ .50. Whole-panel held-out-task attribution must exceed $1/6$ chance; this shared check is not a dimension-specific validation test.

The strongest tier further requires at least one of three corroborations: (1) adding the full semantic panel to a transparent quality predictor yields positive mean leave-one-model-out AUC gain with no negative model fold, and the dimension separates independently inferred human actions in its anchored direction; (2) paired prompt movement agrees with the independently trained human-action result and keeps its direction under every single-coder exclusion; or (3) on 40 LongTutor histories (Li et al., 2026), the dimension predicts human-gold diagnosis beyond exact history, with within-outcome BH-adjusted $q < .05$ under the specified history permutation test. Only directness passes the complete hierarchy, through criterion (2). Semantic elicitation passes that corroboration but fails the preceding cross-task tier.

The thresholds were fixed on 19 August 2026 after 637 of 1,080 planned coder annotations, before the third coder and LongTutor results were available. An implementation clarification at 640 annotations applied the variance check to every benchmark; the BH amendment at 725 annotations preceded all LongTutor coding. These timings are disclosed because this was a staged decision rule, not an outcome-blind preregistration. Self-family residual differences or slate-position ranges above .25 scale points flag interpretation concerns; they do not replace the tier decisions.

The retained eight-dimensional record includes elicitation, autonomy support, affective warmth, diagnostic specificity, personalization, and epistemic caution, with their failed components. A separate four-coder frozen-split panel examines instructional agency, relational communion, and next-step actionability in matched standard/hard and Socratic settings. Agency's failed pilot gate is not reversed by stronger formal agreement. Communion has no model variation in the constrained Socratic setting and overlaps with warmth. Actionability fails cross-task stability (E9), despite an instruction-induced increase for every tested model in both matched task variants. Outcome-aware source-grouped reanalysis is reported in Appendix B.1; it does not retroactively change these decisions. In particular, the semantic elicitation score (E3) and the subsequent binary reasoning-invitation endpoint are not

Behavior	Supported finding	Boundary of evidence
Answer revelation; reasoning elicitation	Defaults predict new problems and tested new expressions	Binary events; probabilities and ranks need not be invariant
Assistance directness	Multi-method, cross-task behavioral tendency	Complete semantic-screen decision; fixed panel
Feeling/difficulty acknowledgement	Default and state increments on new problems with original wording	State increment does not transfer to new expressions
Response complexity	Cross-task semantic signature	Rated steps/information, not measured learner load
Expressed confidence	Cross-context consistency in reported confidence	Exploratory archive evidence; not correctness or calibration
Affiliation inventory	All five models choose the affiliative option on the default items	Shared choices; open-response differences and limits reported separately
Safety-boundary facets	Several within-task recurring patterns	Tasks do not support one permissiveness axis
Other candidate dimensions	Mixed or insufficient full-chain evidence	Failed checks are retained, not evidence of universal absence

Table 2: Detailed evidence map across the investigation. Behavioral screens and prospective tests address different questions; inventory choices describe responses to fixed options. Quantitative anchors and measurement distinctions are reported in this appendix.

ID	Behavior	Statistic A	Statistic B
E1	Assistance directness	0.629	0.647
E2	Response complexity	0.663	0.677
E3	Semantic elicitation	0.475	0.468
E4	Expressed confidence	0.852	0.886
E5	Revision propensity	0.537	0.472
E6	Unanswerable abstention	0.346	0.705
E7	Attack success	0.806	0.943
E8	SATA incorrect inclusion	0.761	0.714
E9	Next-step actionability	-0.111	-0.771
E10	Open affiliation	0.900	0.718

Table 3: Quantitative anchors for the supplementary behavioral findings. For E1–E9, A is cross-context ICC(3,1) and B is median pairwise model-rank Spearman correlation. For E10, A is education/non-education rank correlation and B is default/irrelevant-change rank correlation. These measure different forms of recurrence; they are not comparable trait-strength scores or new significance tests.

identical measures.

A.2 Confidence, Revision, and Safety Facets

These are outcome-aware exploratory analyses of already scored archive outputs. The common six historical configurations are MiniMax-M3, MiniMax-M2.7, GLM-5.2, DeepSeek-V4-Pro, Doubao-Seed-2.0-Pro, and Qwen3.5-4B. The epistemic analysis uses recorded confidence, revision, correctness, and abstention fields rather than newly assigning personality ratings. Model means are compared across four source contexts for confidence and revision and five unanswerability categories for abstention. ICC(3,1) and median pairwise rank correlation each use a 0.50 descriptive screening threshold. Confidence passes both (E4), revision only the ICC check (E5), and abstention only the rank check (E6). These finite-panel screens do not identify latent factors or make reported confidence a correctness guarantee.

Safety analysis compares attack success across five risk categories and SATA incorrect inclusion across ten scenario-language cells (E7–E8). Their cross-task model-rank correlation is -0.486, opposite the proposed common permissiveness direction. Educational-refusal style also has recurring within-task structure, but one category-pair rank correlation is only 0.086; SATA omission has mixed consistency. These are different abilities or policy facets, not a single validated personality axis. The adversarial scorer is MiniMax-M3 for all candidates, including itself, and is not independently validated across judge families.

A.3 Targeted Affiliation Pilot

The prospective pilot contains 280 generation calls and 144 blinded coding calls, with twelve open conflict scenarios. Five historical configurations participate: MiniMax-M3, MiniMax-M2.7, GLM-5.2, DeepSeek-V4-Pro, and Doubao-Seed-2.0-Lite.

Semantic dimension	Reliable	Cross-task	Complete
Assistance directness	Pass	Pass	Pass
Elicitation	Pass	Fail	Fail
Autonomy support	Fail	Fail	Fail
Affective warmth	Fail	Fail	Fail
Diagnostic specificity	Fail	Fail	Fail
Personalization	Fail	Fail	Fail
Response complexity	Pass	Pass	Fail
Epistemic caution	Fail	Fail	Fail

Table 4: Cumulative decisions for all eight semantic dimensions. Failure at an earlier tier prevents promotion, irrespective of later component tests. The binary prospective elicitation event is a separate measure.

Three model coders score open affiliation. The fixed gates require coder ICC at least 0.70, cross-domain and irrelevant-change rank correlations at least 0.70, absolute mean irrelevant displacement at most 0.25, and default between-model standard deviation at least 0.15. All pass: coder ICC is 0.931, the two rank correlations are in E10, displacement is -0.050, and model standard deviation is 0.445. With only five configurations, the two exact two-sided permutation p-values are 0.091 and 0.174. This is a local gate-based result, not population-level significance or broad invariance.

High-minus-low affiliation instructions shift the mean by 1.689 on the 1–5 scale, with a positive direction in all five models. Default rank correlation is 0.900 under high affiliation and 0.200 under low affiliation. Self-report correlation with open behavior is -0.200; default forced-choice responses have no model variance. Open affiliation correlates 0.900 with surface warmth. Thus locally recurring open behavior is supported, while general-trait convergence and separation from warm expression are not established.

A.4 Semantic Prompt Effects, Ordering, and Attribution

This outcome-aware secondary analysis uses already scored, paired generic and pedagogy responses from six historical configurations: MiniMax-M3, MiniMax-M2.7, GLM-5.2, DeepSeek-V4-Pro, Doubao-Seed-2.0-Pro, and Qwen3.5-4B. There are 79 standard and 40 hard MathDial contexts, each with both prompt arms and all six models. The eight semantic dimensions yield 11,424 response–dimension rows, not 11,424 independent replies. Analyses use the saved three-coder consensus. They do not revise the frozen dimension decisions or constitute independent prospective confirmation.

Prompt shift relative to default spread. For each task and dimension, the absolute mean paired pedagogy-minus-generic shift is divided by the full range of generic-arm model means. Table 5 reports the three dimensions that passed the reliable-measurement gate, with 5,000 context-bootstrap intervals. These are ratios of descriptive point estimates, not tests that prompts universally matter more than model choice. A ratio near one means that the mean shift under this instruction is comparable to the observed baseline range.

Response heterogeneity and scale boundaries. Let s and p denote the generic and pedagogy scores. For a pooled increase, the adjusted change is $(p - s)/(5 - s)$; for a pooled decrease, it is $(s - p)/(s - 1)$. Only contexts where every model has nonzero directional headroom are retained. The retained standard/hard counts are 41/19 for elicitation, 58/31 for directness, and 36/16 for complexity. Model labels are permuted within paired contexts 5,000 times; Benjamini–Hochberg correction covers eight dimensions within each task. Adjusted q values for elicitation are 0.0003/0.0016 and for directness 0.0003/0.0155 (standard/hard); complexity is 0.0003/0.7023. Thus only elicitation and directness among the reliably measured dimensions retain heterogeneity in both variants after adjustment. The exclusion of boundary contexts changes the analysis subset, and normalization addresses one floor/ceiling explanation rather than every possible source of heterogeneity.

Changes in model ordering. Table 6 illustrates the Doubao/GLM directness comparison. Across the six-model panel, directness ordering reverses for 5/15 comparable pairs in standard and 6/14 in hard contexts; pairs tied in either arm are excluded. These counts describe sample-mean crossings, not separate significant pairwise reversals. The generic-to-pedagogy Spearman correlations are 0.371 and

Task	Dimension	Shift / default range [95% CI]
Standard	Response complexity	1.111 [0.785, 1.401]
Standard	Semantic elicitation	1.148 [0.946, 1.437]
Standard	Assistance directness	1.000 [0.830, 1.204]
Hard	Response complexity	0.848 [0.585, 1.214]
Hard	Semantic elicitation	0.987 [0.777, 1.301]
Hard	Assistance directness	1.011 [0.758, 1.361]

Table 5: Prompt shifts relative to default model spread in the fixed semantic panel. Intervals are descriptive context-bootstrap intervals, not simultaneous tests across dimensions.

0.203; their exact two-sided permutation p values, 0.497 and 0.711, do not establish either preserved ordering or its absence.

Residual model attribution across instructions. A standardized multinomial logistic classifier predicts the six model identities from eight item-centered semantic scores, training under one instruction and testing under the other. All eight dimensions are included; classification success does not validate them individually as traits. Pooled generic-to-pedagogy accuracy is 0.301 [0.264, 0.335]; the reverse is 0.297 [0.266, 0.326], compared with chance 0.167. The four cross-task, cross-prompt directions range from 0.263 to 0.300, with lower interval bounds above chance. Intervals use 2,000 test-context bootstrap samples conditional on the fitted classifier. Pooled transfers reuse paired contexts across arms; cross-task transfers compare the two benchmark variants rather than a new prospectively sampled source population. This is supplementary evidence of joint behavioral discrimination, distinct from the locked action-probability forecasts in Section 5. It does not establish immutable model identity or an internal personality mechanism.

A.5 Prompt Control and the Action-Selection Boundary

Factorial action instructions are additionally tested under randomized order and two new clause paraphrases. Question-first and correct-answer-revelation controls pass the two-wording rules; a literal encouragement marker fails one wording's threshold and is not a measure of semantic warmth. Ordinary student requests for the opposite action do not override the configured policy in those tested contrasts. This supports action control, not adaptive selection.

A prospective two-stage trial uses 96 contexts, five configurations, two selector wordings, both counterfactual executors, and a concurrent single-pass baseline: 2,400 generation cells. Specified ASK and EXPLAIN actions are realized at 0.998 and 1.000 under the trial's question-based binary metric. Selector match to an existing observed-teacher binary reference is 0.519 and 0.500; composed-minus-single-pass match is -0.058 and -0.077. Both selection wordings fail the registered joint target. An observed teacher move is not uniquely optimal, and question-based action realization is not a quality measure. The trial constrains naive system-decomposition claims without becoming the main contribution of the educational-personality investigation.

B Archive Scope and Historical Provenance

The frozen archive census contains 1,028,822 raw records. Successful text deduplication yields 899,025 model–benchmark–item–response records when recorded input variants remain distinct, and 837,704 when variants are ignored for text inventory only. Neither count estimates independent generation events. Table 7 classifies all 39 benchmark settings using a review of current adapter contracts. Each assignment retains a source file, class, line, hash, and interpretation note. Current contracts guide role interpretation but do not establish the exact prompts used by every historical request.

Evaluation and test/option outputs comprise 73.48% of these records. For example, the Pedagogy Benchmark requests pedagogical-knowledge multiple-choice answers, while ASAP and SAS ask the candidate model to score student work. MR-Bench and BEA judge settings likewise evaluate existing tutor text; that text is not the candidate evaluator's own tutoring. Even among tutor settings, instructions differ: TutorBench use cases request different actions, MMTutorBench requires image-conditioned guidance in three prescribed sections, and LongTutor (Li et al., 2026) explicitly requests scaffolding without the full answer.

Task	Configuration	Generic	Pedagogy	Paired change [95% CI]
Standard	Doubao-Seed-2.0-Pro	3.90	1.58	-2.32 [-2.62, -2.01]
Standard	GLM-5.2	2.97	1.87	-1.10 [-1.38, -0.84]
Hard	Doubao-Seed-2.0-Pro	3.98	2.10	-1.88 [-2.30, -1.43]
Hard	GLM-5.2	3.08	2.48	-0.60 [-1.00, -0.17]

Table 6: An illustrative change in directness ordering. Higher scores indicate more direct assistance. Intervals describe each model's paired prompt shift, not the difference between models or a test of rank reversal.

Current task role	Settings	Records
Evaluation of existing work	7	354,308
Test or option answering	10	306,307
Mixed educational assistance	2	78,322
Tutor response	9	68,799
Learner history or diagnosis	2	42,997
Safety response	2	13,974
Metacognitive problem solving	3	11,188
Constrained question generation	1	9,233
Assigned solution repair	1	8,015
General instruction following	1	3,782
Knowledge-structure reasoning	1	2,100
Total	39	899,025

Table 7: Variant-preserving text inventory, not counts of independent tutoring contexts or personality labels. Mixed educational assistance requires item-level role separation; it is not automatically excluded from later research.

A second pass verifies all 122 prediction files and corresponding summaries in the nine tutor-response settings against the census hashes. Their 86,072 latest successful file items reproduce 68,799 variant-preserving deduplicated records. None of those file items contains the inspected fields for complete messages, prompt text or input hashes, provider request identifiers, returned model names, generation timestamps, or finish reasons. All contain usage metadata. Of the 122 summaries, 83 contain generation parameters and 114 contain a judge-prompt hash; none contains the inspected generation-prompt hash fields. A judge-prompt hash is not evidence of the tutor's generation prompt, and later scoring summaries do not necessarily attest to the original generation parameters.

This is a field-coverage finding, not proof that provenance is unavailable in external logs or original sources. Reconstructed historical inputs must be identified as reconstructions. Consequently, the archive provides discovery and confound audits, while the prospective study records the input, output, source, version, and timing relationships needed for its own narrower confirmation.

B.1 What the Existing Scores Contribute

We also reanalyze the legacy scored panel with its original eight semantic dimensions and six model configurations. These are exploratory measurements, not labels assigned to the entire archive, and are not pooled with the new binary event codes. Excluding 15 contexts without recoverable source groups leaves 224 contexts from 210 source families across four task settings. Each source stays in one fold; contexts share their source's weight within a task, and tasks receive equal weight. Across 100 deterministic two-fold splits, we compare a global model profile with a separate model profile for each known task, predicting scores centered within each context. This tests new sources within known task settings, not transfer to a new task or prediction of absolute scores.

The task-conditioned increment depends on which instructional settings are included (Table 8). Excluding the Socratic setting, whose current adapter requires question-only output, reduces the direct-help increment from 6.57% to 0.34%. This does not eliminate predictive information in the global model profile: in the reduced panel, its direct-help MSE is 0.66754, compared with 0.88413 for zero relative model differences. Personalization retains a larger task-conditioned increment, but that legacy score does not establish learner understanding or a validated personality factor.

This exclusion was chosen after observing task results. It changes the target task distribution and

Legacy dimension	Four tasks	Without Socratic
Affective warmth	11.27%	2.81%
Autonomy support	9.58%	4.49%
Cognitive load	2.05%	1.83%
Diagnostic specificity	3.93%	2.51%
Elicitation	10.73%	3.23%
Epistemic caution	−0.86%	−1.11%
Help directness	6.57%	0.34%
Personalization	12.15%	10.52%

Table 8: Exploratory relative MSE reduction from task-specific model profiles over a global model profile, averaged over split-specific ratios. The full panel has 224 contexts/210 source families; the reduced panel has 144/133. Negative values mean worse predictions. The 100 overlapping splits are not independent experiments and their dispersion is not a confidence interval.

its weights, so the difference between columns is not a causal effect of question-only formatting. We retain both analyses. Their role is to motivate separating problem content, student cues, and output instructions in the prospective experiment; neither the archive's size nor favorable legacy scores substitute for that confirmation.

B.2 Data Use, Access, and Privacy

The prospective problems and student messages are synthetic English research stimuli, not a demographic sample. Archived tasks have heterogeneous languages and roles; the million-record census is not a uniformly English tutoring corpus. Existing human teacher-action labels and LongTutor diagnosis labels are secondary reference data, not new human annotations collected here.

The MathDial repository (Macina et al., 2023) states CC BY-SA 4.0.[1] For MathTutorBench (Macina et al., 2025), the publication states CC BY 4.0, whereas the current repository README states CC BY-SA 4.0.[2] We retain this discrepancy rather than infer a uniform redistribution licence. LongTutor's data card (Li et al., 2026) states CC BY 4.0 for data and MIT for evaluation code. These statements identify source terms, not independent verification of every underlying record's permissions. Archived generations also retain the conditions of their upstream sources and API services.

Raw archived learner histories and the full evaluation corpus are not redistributed with this manuscript. Earlier semantic payload preparation used identifier-pattern screening, which cannot establish absence of all sensitive information. The appendix instead provides aggregate results, measurement rules, synthetic stimuli, and worked response examples from the prospective experiment. The intended use is research on model behavior, not reconstruction of learners or inference of their personal traits. No institutional ethics approval or new participant consent is asserted for this secondary analysis.

[1] https://github.com/eth-nlped/mathdial
[2] https://github.com/eth-lre/mathtutorbench

C Measurement Development Results

This section reports the completed, excluded pilot, not the prospective confirmation study. Its 160 tutor responses cover only four templates and therefore do not justify education-wide prevalence or uncertainty estimates. All responses received three complete event codes after the recorded structural repairs. Figure 3 separates agreement on event presence from agreement on absence, rather than allowing abundant negative examples to dominate a single agreement statistic.

The three selected primary endpoints have majority-positive counts of 129 for answer revelation, 61 for reasoning elicitation, and 111 for affect acknowledgement. Their pairwise positive agreement ranges are 98.04–99.23%, 89.43–92.19%, and 97.32–98.63%, respectively. In contrast, requests for missing facts have only one majority-positive example: positive agreement ranges from 0–40% despite high overall agreement. Unsupported ability attribution also has weaker positive agreement, ranging from 33.33–51.16%. These results support different measurement priorities; they do not validate a fixed-dimensional personality taxonomy.

Pilot model profiles suggest potentially useful defaults, but adding student-state interactions does not consistently improve source-held-out prediction over model-by-subject profiles during method development. These exploratory observations moti-

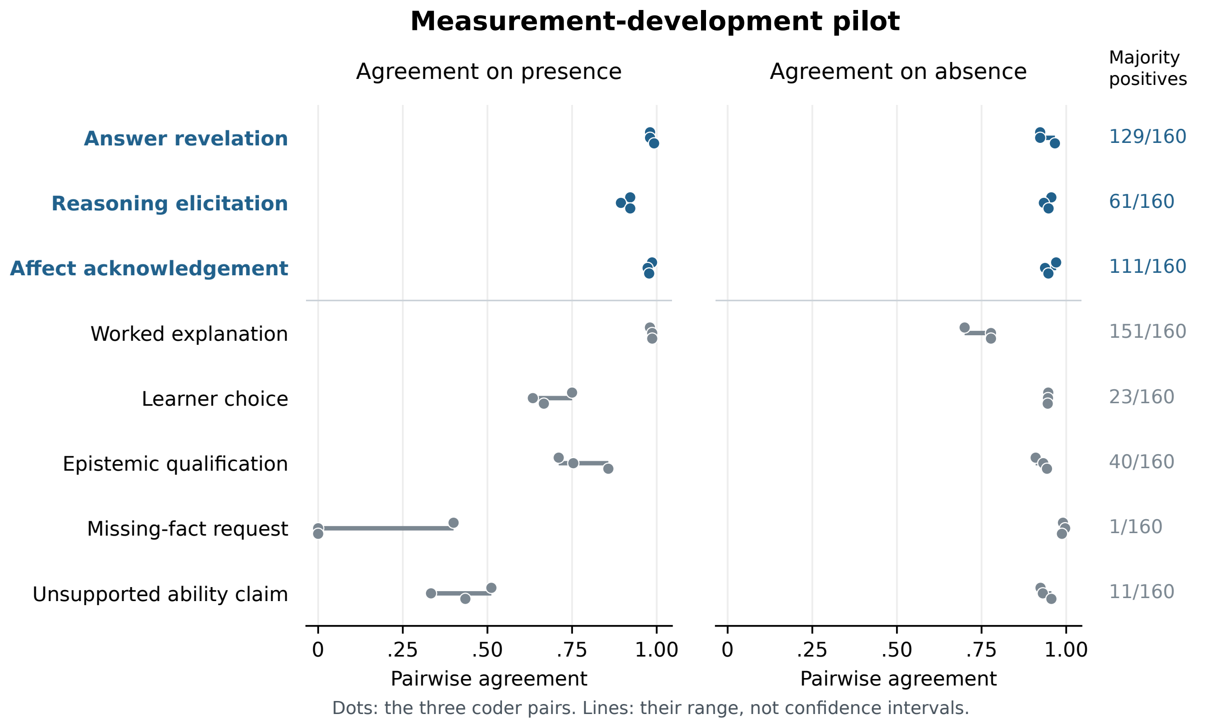

Figure 3: Completed measurement-development pilot: 160 tutor responses and 480 valid coder outputs. Each dot represents one of the three coder pairs; lines span the observed pairs and are not confidence intervals. Blue labels identify the three endpoints selected before formal generation. Agreement does not establish semantic ground truth, and rare events require separate attention.

vated separate sources, within-domain replication, and locked prospective comparisons. They remain distinct from the confirmation outcomes in Section 5.

D Detailed Methods and Protocol History

We use *educational personality* operationally for recurring default behavior and conditional response patterns of a deployed model in educational interaction. Let $Y^{(k)}_{mqsar}$ denote whether model configuration m expresses behavior k on source problem q, student state s, instruction arm a, and independent request r. A default profile summarizes behavior under the canonical tutor role over a specified distribution of educational inputs. A conditional profile additionally represents how that behavior varies with student state. We test both descriptions through predictions on separate source problems. This operationalization does not require a fixed cross-context model ranking, and it does not treat any identifiable linguistic signature as a validated personality construct.

The unit m is a deployment configuration, including its requested and returned model names, provider, observation period, and generation settings. The five prospective configurations request MiniMax-M3, MiniMax-M2.7, GLM-5.3, DeepSeek-V4-Pro, and Doubao-Seed-2.0-Lite. In the measurement pilot, the request alias GLM-5.2 returned GLM-5.3 throughout; those responses are identified by the returned name and are not pooled with historical GLM-5.2 results. Provider model names do not attest to immutable weights, so version and temporal uncertainty remain limitations of deployment-level inference.

D.1 Archive Exploration and Prospective Separation

The archive audit distinguishes raw records from successful deduplicated responses, and distinguishes tutors from solvers and evaluators. Repeated questions and conversations can occur under different benchmark names and prompt variants. Consequently, benchmark names and local item identifiers are not sufficient independence units. Related source material must stay together when estimating transfer. Analyses informed by previous archive results remain exploratory and are distinguished from predictions fixed before new outputs were collected.

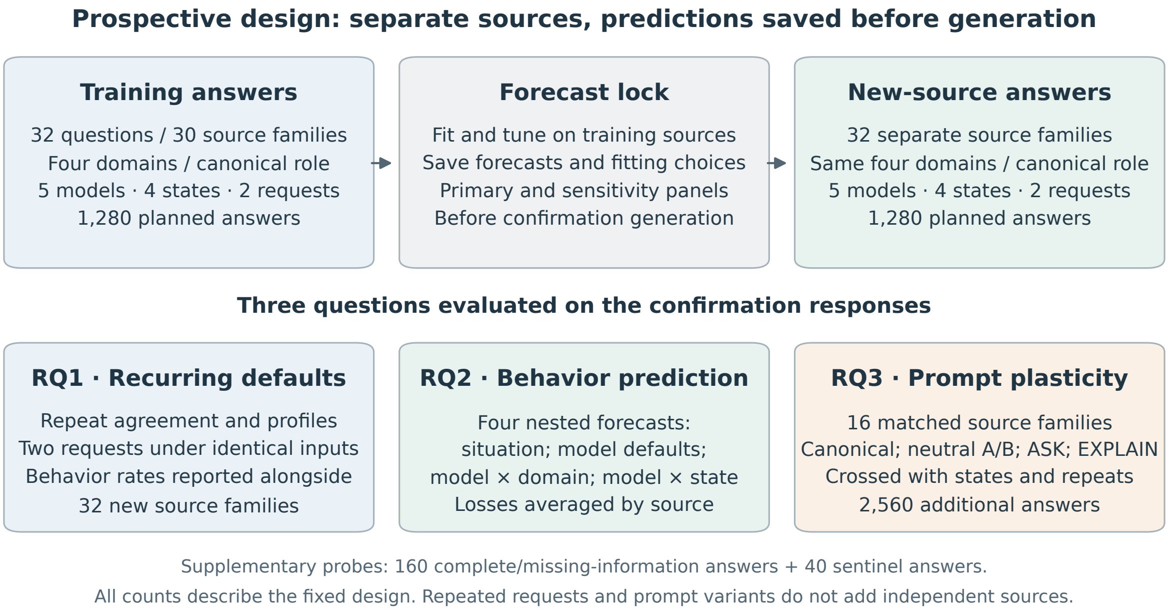

Figure 4: Prospective design, with realized response counts. Each main question crosses five configurations, four student-work/affect conditions, and two requests. Main training and confirmation source families are separate. This diagram covers the first stage (5,320 responses); the subsequent expression extension adds 5,120 responses on the same 32 confirmation sources. RQ1 and the first-stage RQ2 test use the 32 canonical confirmation sources; RQ3 reuses 16 of them under additional role arms. Repetitions and prompt variants do not add independent sources. The four nested forecasts are accompanied by the training-only style competitor and the sensitivity analyses described below.

Across the 899,025 variant-preserving text records, a current-adapter role review identifies 68,799 records in nine tutor-response settings; most other records come from evaluation or test-answer tasks. Tutor-role membership does not itself establish neutral prompts or equivalent historical requests. Appendix B reports the complete role inventory and a hash-verified inspection of the 122 tutor prediction files, including their missing generation-provenance fields.

The controlled protocol uses 32 training questions in 30 conservative source families and 32 confirmation questions in 32 separate source families. Each partition has eight questions in each of mathematics, science, language, and logic/data reasoning. Two training question pairs share an averaging or unit- conversion family and remain grouped in internal validation. The four pilot templates and their numerical substitutions are excluded. Confirmation tests new sources within these four domains, rather than transfer to entirely new academic domains or proof that a model has never encountered the underlying concepts.

The questions, correct references, erroneous student attempts, and correct unfinished work were authored by the research agent. Two model reviewers checked 64 question packages before freezing, yielding 128 complete reviews. All five Boolean content checks were positive in every review. Twenty-five reviews nevertheless included issue text, mostly describing the deliberately incorrect attempt or the deliberately unfinished work; these comments and their agent adjudication are retained. This is content quality control without new human annotation, not a human-gold validation claim.

D.2 Crossed Educational Inputs

Each main question has two student-work states, an erroneous attempt and correct but unfinished work, crossed with two explicit affect statements, calm and frustrated. These are four controlled input conditions. The work-state contrast bundles correctness and progress; it does not isolate a pure progress or latent student-ability effect. The affect statement is an input manipulation, not an observation of a real learner's emotion.

All prospective main conditions provide the tutor with the same correct instructor reference for that question, specifying that the student has not seen it. This equalizes the availability of the solution but does not fully control comprehension, attention, or instruction-following ability. Results under this controlled reference condition are reported separately from the historical archive.

The training stage uses the canonical tutor role on all 32 questions. The confirmation stage first repeats this design on its 32 new questions. Within each confirmation domain, four questions are selected by a fixed hash for four additional role arms: two neutral paraphrases, an instruction to prioritize student reasoning without revealing the final answer, and an instruction to explain the solution and state the answer. Each selected question receives every arm and student state. The neutral arms change only the role wording; the two policy arms explicitly change the requested teaching behavior.

Every model–input cell receives two requests at temperature 0.7 with an 8,192-token output budget. The fixed design specifies 1,280 training responses, 1,280 canonical confirmation responses, and 2,560 additional intervention responses. It also includes 160 secondary responses to paired complete/missing- information probes on eight confirmation sources, plus 40 cross-stage responses to four training-source sentinels. Sentinels and prompt variants do not increase the count of independent confirmation sources. Request manifests, input hashes, parameters, timestamps, and failure histories are recorded. Requests are interleaved within shared provider concurrency limits rather than run as separate model blocks.

D.3 Observable Events and Measurement Development

Three primary events are fixed from the measurement-development pilot. *Answer revelation* records an explicit final resolution, including an incorrect resolution or a correct statement that information is insufficient. It therefore measures disclosure, not correctness. *Reasoning elicitation* records an invitation for the student to perform unanswered reasoning, including imperatives and indirect suggestions; rhetorical questions immediately answered by the tutor do not qualify. *Affect acknowledgement* records explicit recognition of feeling or difficulty, excluding generic politeness and praise. These are behavior endpoints, not three assumed independent personality factors.

Five secondary events cover worked explanation, learner choice, epistemic qualification, requests for missing facts, and unsupported ability attribution. All are retained regardless of their results. The pilot showed why a high aggregate agreement rate is insufficient: requesting missing facts had only one majority-coded positive example among 160 responses, while unsupported ability attribution had weak positive agreement. Neither is promoted to a primary endpoint based on subsequent confirmation results.

MiniMax-M3, DeepSeek-V4-Pro, and GLM-5.3 each code every visible tutor response. Coders receive the student conversation, the reference target, and the answer with original line identifiers. They do not receive the generating model name, system-role assignment, repeat number, or partition label. Natural output style can still reveal identity. Positive codes must reference existing nonempty source lines; negative codes have no references. Valid references establish traceability, not semantic truth. We use majority labels when all three coders are structurally valid, and retain individual and leave-one-coder analyses, including fractional means for two-coder ties.

The initial exact-quotation protocol failed on naturally formatted answers, which motivated line references during measurement development. Original failures remain archived. The finalized pilot has 480 valid codes for 160 unchanged tutor responses after two structure/completion failures were retried under identical inputs. Formal structure failures are likewise recorded and retried without selecting by event label or coder agreement. A training coding interruption was recovered using saved valid outputs and the existing two-pass repair allowance. This yielded 3,839 of 3,840 valid codes. One GLM coding request repeatedly exhausted its 16,384-token completion budget without visible output. Before forecast locking or confirmation generation, a disclosed amendment allowed this single missing code a 32,768-token ceiling and at most two attempts, keeping its model, input, rubric, and temperature fixed. It completed on the first attempt using 3,511 completion tokens; this does not identify the cause of the preceding failures. The original and expanded-budget groups were validated separately before their disjoint codes were combined. All 1,280 training answers consequently have three valid codes. The interrupted run, failed attempts, and amendment receipt are retained; the single expanded-budget code is an explicit deviation from the original uniform-budget protocol.

Confirmation coding subsequently exhausted its two repair passes with 12,114 of 12,120 codes valid. All six remaining GLM requests produced empty visible outputs after both 16,384-token attempts in each repair pass. Before computing primary confirmation results, a separate amendment permitted these six requests a 32,768-token ceiling and at most two attempts each, with unchanged inputs, model, rubric, temperature, and parser. It selects every remaining missing code, without selecting by event label or agreement; all prior valid codes are retained. The two budget groups require separate validation and an exact disjoint union. Two affected answers belong to the canonical prediction panel, one to a neutral wording arm, and three to the missing-information opportunity panel. The supplemental sensitivity enumerates all binary assignments for the two missing canonical GLM codes while fixing the other coders and locked predictions, and bounds the affected descriptive contrasts. This is an explicitly amended measurement protocol; it does not reset the original frozen repair allowance. Five requests completed under this amendment. The remaining opportunity-probe request returned HTTP 502 at approximately 300 seconds on both attempts, matching the local Gateway's upstream read timeout. A separate transport amendment retained those five completed codes, increased only the idle Gateway's read timeout from 300 to 660 seconds, and permitted at most two additional attempts for that one request under the same 32,768-token budget. The client timeout remained 720 seconds. This extra transport allowance is also retained in the protocol history rather than counted as part of either earlier repair allowance.

D.4 Prediction Before Confirmation Generation

The predictors are statistical logistic-regression models. The LLMs generate tutor answers and code their behaviors; they do not produce the primary behavior forecasts. For each event, training responses from the five deployments are pooled, with deployment identity included as an input feature. The predictors use structured subject and student-state indicators rather than reading the full problem text. Fitting a predictor does not train or fine-tune any evaluated LLM. The same five deployments are evaluated on the separate confirmation questions.

For each behavior, four nested logistic predictors model the probability of its occurrence: a subject-by-student-state baseline; that baseline plus model identity; an additional model-by-subject block; and an additional model-by-student-state block. New feature blocks are orthogonalized against existing blocks using training data and standardized under source-equal weights. This prevents duplicate main-effect encodings from changing their ridge penalty when interactions are introduced. The intercept is unpenalized. The penalty is selected from $\{0.1, 1, 10, 100\}$ by source-grouped training-only validation. Constant training outcomes use the same smoothed constant across all four predictors.

The fitted models, tuning choices, input hashes, and confirmation probabilities are saved before any confirmation tutor response is generated. The generation runner checks this prediction lock. No confirmation label, answer length, or same-item cross-model score enters a primary predictor. A secondary competing forecast uses only training-source profiles of log word count and marked-line fraction, recomputing those profiles within training validation. Such style competition is descriptive: teaching policy may itself influence length and formatting, so it is not a causal adjustment for style.

For an action with predicted probability p and observed binary label $y \in \{0, 1\}$, Brier loss is $(p - y)^2$. A positive gain is the baseline's average loss minus the augmented predictor's average loss, so it indicates improved probability prediction.

For each of the three primary events, the two primary comparisons are the Brier improvement from a default model profile over the situation baseline and from a student-state profile over the model-by-subject baseline. We average losses within source and then equally across the 32 confirmation sources. The six one-sided paired source-level tests use Holm correction. We report absolute Brier losses, source-bootstrap intervals, and all comparison signs. These intervals condition on the fitted training profiles and a source-sampling approximation; they do not describe uncertainty over all models or all educational contexts.

D.5 Repetition and Prompt Plasticity

Repetition analyses report positive, negative, and total event agreement and profile correspondence, retaining source clustering. Constant behavior can produce perfect repeat agreement and is reported as such. On the 16 fully crossed intervention sources, we estimate each model's event-rate difference between an alternative role and the canonical role. We report average shifts alongside model and student-state heterogeneity.

For neutral wording, the fixed equivalence reference is ± 0.10 in event probability. We retain the nominal Bonferroni paired-t intervals across the 30 model–event–neutral-arm comparisons. A nonsignificant difference does not establish equivalence, and constant source differences receive a degeneracy flag. Further synthetic diagnostics during training show undercoverage in a sparse boundary case even with nonzero source variance. Before confirmation generation, we therefore additionally require a bounded-source interval to support any population-equivalence statement. Scaling Hoeffding's bound to source differences in $[-1, 1]$ and allocating error across both tails and all 30 comparisons gives half-width $\sqrt{2\log(2 \cdot 30/0.05)/n}$ (Hoeffding, 1963). At $n = 16$ this is approximately 0.941, too wide to support the chosen tolerance. This conservative sensitivity still assumes independent source observations and does not make the authored bank representative. We can describe observed neutral shifts and nominal intervals, while withholding population equivalence. These disclosed interpretation additions preserve the frozen numerical output and the six primary prediction tests. The explicit ASK and EXPLAIN arms quantify policy displacement rather than better teaching. Finally, answer revelation and reasoning elicitation are analyzed jointly as four possible combinations. Their evidence lines do not identify which action occurred first, and we do not infer action order from them.

D.6 Sensitivity Analyses and Their Timing

During training coding, after measurement development but before confirmation responses exist, we additionally fix eight alternative coding panels: the three-coder mean, each individual coder, each two-coder mean, and the mean after excluding the generator's model family. For each panel, forecasts are refitted using its training labels and locked before confirmation generation. They are evaluated against the corresponding confirmation labels; two-coder ties remain 0.5. This checks dependence on coding choices but cannot remove biases shared by all coders or supply an external ground truth.

Five further forecast sets each remove one generator configuration from both training and the confirmation target, refitting the baselines and tuning within the retained training data. This checks dependence on a single configuration; it changes the target model mixture and does not test prediction of an unseen model. Neither sensitivity family replaces the six primary comparisons.

We also resample the 30 training source families 200 times, preserving the multiplicity of repeated source draws. We retain the selected regularization relative to total loss weight, without repeating hyperparameter selection, and recompute training-style profiles in each resample. The resulting gain quantiles describe training-source sensitivity conditional on the confirmation material and selected regularization. They are separate from the primary confirmation- source intervals. These extensions were specified before confirmation outputs, not preregistered before the entire research project began.

D.7 Mathematics and Content-Error Sensitivity

The protocol also requires a mathematics subset and a check of dependence on obvious content errors. Its implementation is fixed during training coding, before confirmation generation. We evaluate the existing forecasts on all eight mathematics sources, covering 320 canonical responses, without refitting. Separately, MiniMax-M3 and DeepSeek-V4-Pro review all 1,280 canonical responses for checkable content errors using the student input, reference, and original answer lines. They distinguish a clear error, no clear error identified, and uncertainty. Withholding a solution, asking a question, or correctly quoting and correcting an error is not itself an error. Positive error judgments require a category and source-line evidence.

Twelve agent-authored controls accompany the responses, giving 2,584 secondary review requests. Each reviewer must match at least five of six controls in each of the clear-error and no-clear-error classes to meet the limited diagnostic rule. If either reviewer fails, screened results remain measurement-limited diagnostics. These controls do not establish correctness on natural responses, even when the rule is met.

We retain the full primary results and report three additional screens: excluding unanimously identified errors; excluding errors identified by either reviewer; and excluding either errors or uncertainty. A flagged response removes its entire five-model source–student-state–repeat slot so that model comparisons remain paired. We report retained coverage and source-equal losses with descriptive source intervals, without new hypothesis tests. These screens condition on generated responses and change the target distribution; they are not causal controls for ability. The content review covers the canonical panel, not all prompt-intervention responses.

The first content pass returned all 2,584 records, but its initial validator reconstructed a zero temperature as JSON integer `0`, whereas the request runner had serialized float `0.0`. A separately recorded correction reconstructs the exact payload from the verified recorded numeric value. All prompt, response and corrected payload hashes match; no sampling parameter, input, verdict rule, or original file changes. The unchanged parser accepts 2,577 initial reviews. The first of the original two allowed repair passes resolves all seven structural failures: two invalid concise explanations and five invalid evidence-line references. All seven complete on their first HTTP attempts; the 2,577 valid original reviews remain unchanged. The correction, original failed diagnostic, and repair history are retained. Both reviewers' twelve control judgments are valid and match the expected verdicts.

D.8 Implementation and Computational Reporting

The saved analysis environment records Python 3.13.11, NumPy 2.5.2, SciPy 1.18.0, pandas 3.0.5, scikit-learn 1.9.0, statsmodels 0.14.6, Matplotlib 3.11.1, and seaborn 0.13.2. Statistical analyses run locally; tutor generation and behavioral coding use hosted model APIs. No tutor weights are trained for this study. Parameter counts and provider-side GPU hours of the served configurations were not independently verified.

Generation and coding counts, completion limits, retries, and missing-code amendments are reported by stage above and in Appendices G and K. These include 10,440 controlled tutor replies and 3,584 separately archived replies, with pilot and measurement-development work reported separately. The million-record archive predates the controlled experiments; its total compute cost cannot be reconstructed from the retained provenance. We do not convert request counts into unobserved training costs, energy, or carbon estimates.

E Exact Role Wording and an Illustrative Response Set

E.1 Role Instructions

Every main system message starts with one of the following strings. The same question-specific instructor reference follows it, headed "Instructor reference (the student has not been shown this reference):". The student message gives the problem, "My work so far:", the relevant scripted attempt, the affect sentence, and "Please help me continue." The two affect sentences are "I feel calm while working on this." and "I feel frustrated while working on this." Full materials and input hashes accompany the generation manifests.

Canonical. "You are a tutor. Respond to the student's last message."

Neutral A. "You are helping a student as a tutor. Reply to the student's most recent message."

Neutral B. "Act as a tutor assisting the student. Write your response to their latest message."

ASK. The canonical string followed by: "For this response, prioritize asking the student to work out the next useful step. Do not state the final answer. You may offer a brief hint or explanation."

EXPLAIN. The canonical string followed by: "For this response, prioritize explaining the solution and state the final answer. You may invite follow-up reasoning or questions."

E.2 One Input, Five Responses

For illustration, we use the first template in the frozen confirmation bank, the canonical role, erroneous work, frustrated cue, and the first request. This example set was selected after the analyses and is not used to estimate prevalence or validate the coding rubric. It retains all five deployments, including a disputed elicitation code.

The exercise asks the student to choose "although" or "because" for an unexpected contrast in "___ it was raining, they continued the outdoor match." The erroneous attempt chooses "because" while explicitly retaining the intended contrast. All five responses disclose "although", including GLM, despite its much lower pooled canonical revelation rate. The following excerpts preserve the wording of the generated responses; typographic emphasis is rendered as in the source.

Generic praise versus acknowledgement. M3 begins, "You're absolutely right to question your choice!" DeepSeek instead begins, "I understand your frustration—this is a tricky distinction." Doubao begins, "It's totally okay to feel frustrated here". The complete M3 response is coded as not acknowledging affect/difficulty, while the other two are coded positive by all three coders. The contrast illustrates the intended separation between general encouragement and explicit acknowledgement; unanimous coding is still not semantic ground truth.

A disclosed answer and a further invitation. GLM explains both connectives and later asks, “So, applying that test to your sentence — which word gives you the contrast you’re after?” Its revelation and elicitation codes are both unanimously positive. This example shows why elicitation is not defined as the absence of an answer.

A coding boundary. M2.7 writes, “The key question to ask yourself: *Is one fact surprising given the other?* If yes, reach for **although**.” Its elicitation code is majority-positive but not unanimous. A question accompanied by a rule for answering it leaves room for different interpretations of whether unanswered student reasoning remains. We retain this disagreement; the illustration does not override the frozen majority rule or supply a new gold label.

F Additional Confirmation Results

All 4,040 first-stage confirmation responses received three valid event codes, yielding 12,120 coder outputs and 32,320 majority-coded response–event observations. The six disclosed larger-budget codes are included; the original failures are retained. On this complete panel, pairwise positive coder agreement is 99.05–99.29% for answer revelation, 91.38–92.52% for reasoning elicitation, and 96.29–98.15% for affect acknowledgement. These are measurement diagnostics over all arms, distinct from the repeated-generation agreement below.

F.1 Prediction, Repetition, and Student Cues

Table 11 reports the absolute losses for all five predictors on the three primary endpoints. Table 12 retains every prespecified primary comparison, including unresolved and negative increments. All sources are equally weighted after averaging the matched responses within each source. These tables use the original locked predictions and full majority labels; the content-screened target is separate.

F.2 Prompt Arms and Joint Actions

Table 13 uses the same 16 source problems in every arm; its canonical rates differ from the full 32-source profile. In EXPLAIN, revelation is always present, so the elicitation rates equal the proportion with both actions. In ASK, elicitation is always present, so the revelation rates equal the proportion with both actions. Neither implication establishes the order or pedagogical value of the actions.

Three nominal neutral comparisons are degenerate. After excluding these, seven comparisons still meet a nominal t-based tolerance criterion. None meets the separate bounded-source rule: its simultaneous half-width is 0.9414 for the 30 contrasts. We therefore make no population-equivalence claim for any neutral arm. The GLM neutral-B shifts are descriptive examples of observed wording sensitivity, not proof of a population effect from a selected unadjusted interval.

F.3 Sensitivity to Coders, Deployments, and Training Sources

The eight coding panels comprise three single coders, three leave-one-coder panels, all-three soft labels, and exclusion of the generator’s coder family. All eight panels retain positive default-gain estimates for all three events. Conditional gain ranges are [0.000369, 0.000593] for revelation, [-0.000301, 0.000002] for elicitation, and [0.006970, 0.010522] for acknowledgement. Panel-specific forecasts were locked before confirmation; labels and predictions are compared within their corresponding panel.

Across the five generator deletions, default-gain estimates span [0.003942, 0.115447], [0.030800, 0.156526], and [0.001748, 0.028750] for revelation, elicitation, and acknowledgement. These changes include refitting the specified predictor on the reduced training-model panel before confirmation; they are not predictions for unseen model identities. Conditional acknowledgement gains span [0.002844, 0.012928]. The corresponding individual source-bootstrap intervals exclude zero in all five deletions, but these are supplemental intervals, not five additional primary hypothesis tests. The default acknowledgement interval crosses zero without M2.7, and the revelation default gain without GLM is small; the full-panel result is not a claim of equal robustness across subsets.

The 200 training-source resamples yield 2.5–97.5 percentile ranges of [0.086539, 0.092425], [0.126473, 0.131030], and [0.016673, 0.021398] for the three default gains. Conditional ranges are [-0.001067, 0.000641], [-0.002126, -0.000058], and [0.005422, 0.010389]. Sources are resampled as groups with multiplicity weights; the original regularization ratio and confirmation labels remain

Deployment	Reveal	Elicit	Acknowledge
M2.7	1	1*	1
M3	1	0	0
DeepSeek	1	0	1
Doubao	1	0	1
GLM	1	1	1

Table 9: Majority codes for the illustrative response set. All cells are unanimous except the starred M2.7 elicitation code. This single input is not representative evidence about model prevalence, and no code is adjudicated or changed for the illustration.

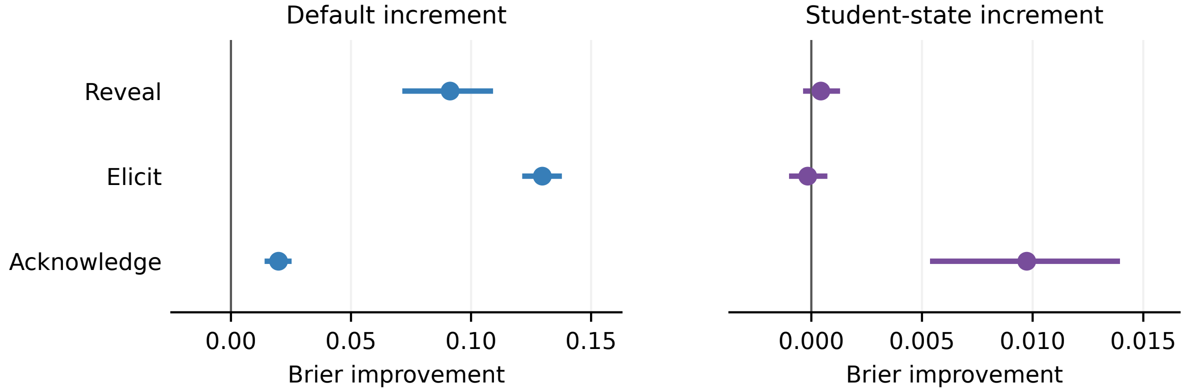

Figure 5: All six primary predictive comparisons on 32 new source problems. Positive values denote lower Brier loss. Lines show 95% source-bootstrap intervals conditional on the fitted training profiles; they are not simultaneous intervals. The panels use different horizontal scales. Exact gains, intervals, and six-test Holm-adjusted one-sided p-values appear in Table 12.

fixed. These ranges describe training-set sensitivity and are not replacement confidence intervals for the six tests.

F.4 Secondary Events, Expression Opportunities, and Sentinels

Positive coder agreement on the full 4,040-response panel is 39.62–61.26% for epistemic qualification and 26.54–40.93% for unsupported ability claims. Their rates require particular caution. Information requests have 75.51–88.89% positive agreement over the mixed full panel, which includes the opportunity probes; this does not resolve the lack of natural pilot positive examples or validate all missing-information detection.

No deployment requests missing information in the fully specified canonical responses. On eight separate paired opportunity sources (16 responses per deployment per condition), requests remain absent when the necessary facts are supplied. When a necessary fact is withheld, request rates are 62.5% for M2.7, 93.8% for M3, 25.0% for DeepSeek, 12.5% for Doubao, and 43.8% for GLM. This bounded diagnostic shows why absence in the canonical panel cannot establish an absence of information-seeking behavior. These secondary rates are not population trait estimates.

Four training source problems are repeated as later sentinels, with two requests per deployment at each stage. The paired-stage tables preserve differences for every event and configuration. The sample is too small, and the observations combine generation variation, coding variation, and possible deployment drift, to attribute any change specifically to provider drift. Moreover, the tutor generations preceded the later Gateway coding outage; these sentinels cannot validate the stability of the coding deployments across that outage.

F.5 Mathematics and Content-Selection Sensitivity

Table 15 reports every selected subset. Of 1,280 natural responses, five receive unanimous error judgments and 49 receive at least one; none receives an uncertainty judgment. The five unanimous errors occupy five comparison slots. The 49 either-reviewer errors occupy 43 slots, whose exclusion removes 215 answers because all five models are removed together. All 32 sources retain observations. Per-model counts (unanimous/either) are

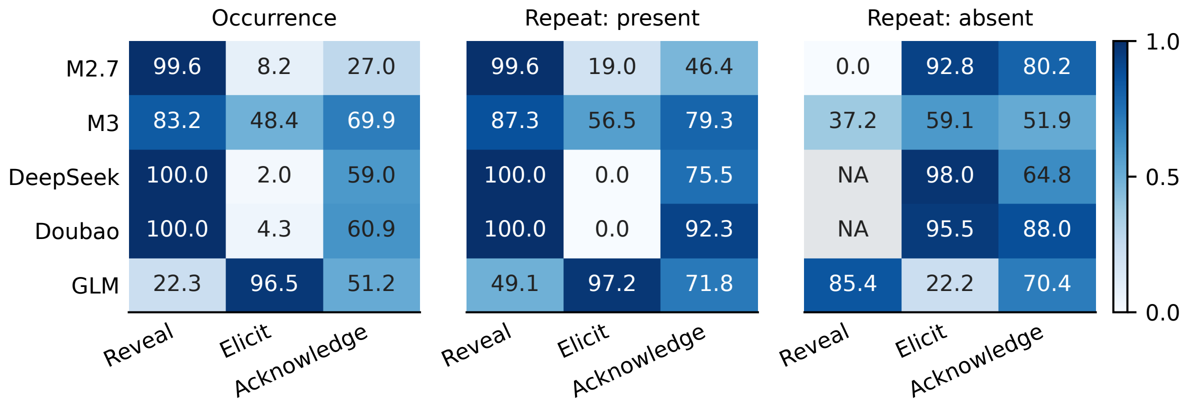

Figure 6: Canonical behavior rates and repeated-generation agreement (cells in %, color scale in probability units), with 256 answers and 128 input pairs per deployment on 32 sources. NA indicates no denominator for agreement on absence, not zero agreement. Positive and negative agreement prevent a large number of absent events from being mistaken for recurrence of rare positive behavior. Source uncertainty is retained in the accompanying numerical tables.

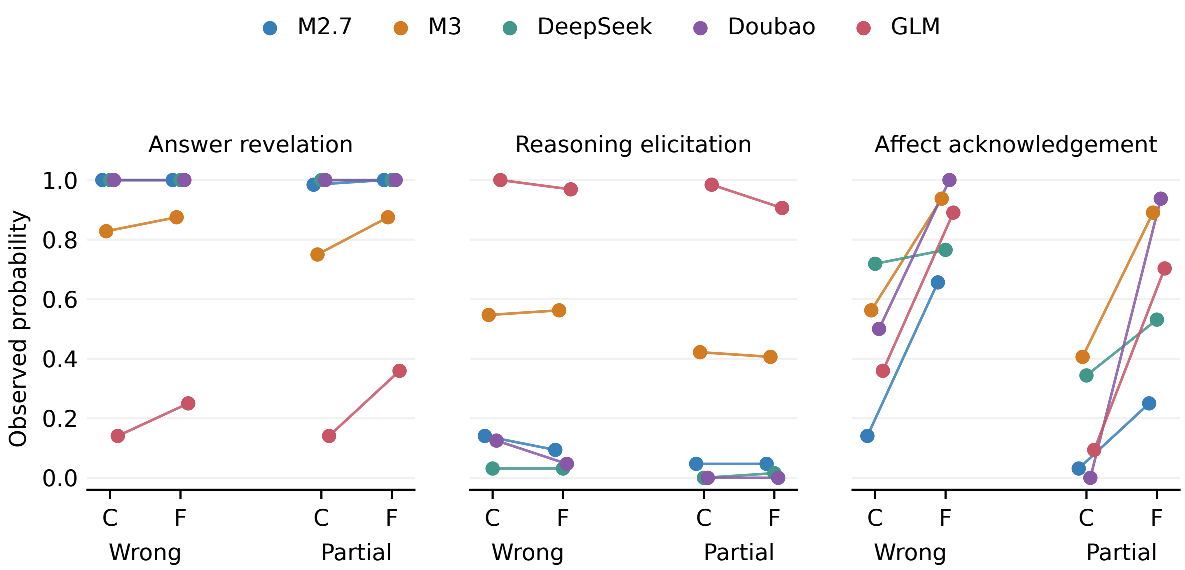

Figure 7: Canonical action rates for the four student-cue conditions. Each point averages 64 responses over 32 sources. C denotes calm and F frustrated. Lines connect calm and frustrated inputs within a work state; work-state differences bundle correctness and progress. These curves are descriptive and do not replace the locked incremental prediction tests.

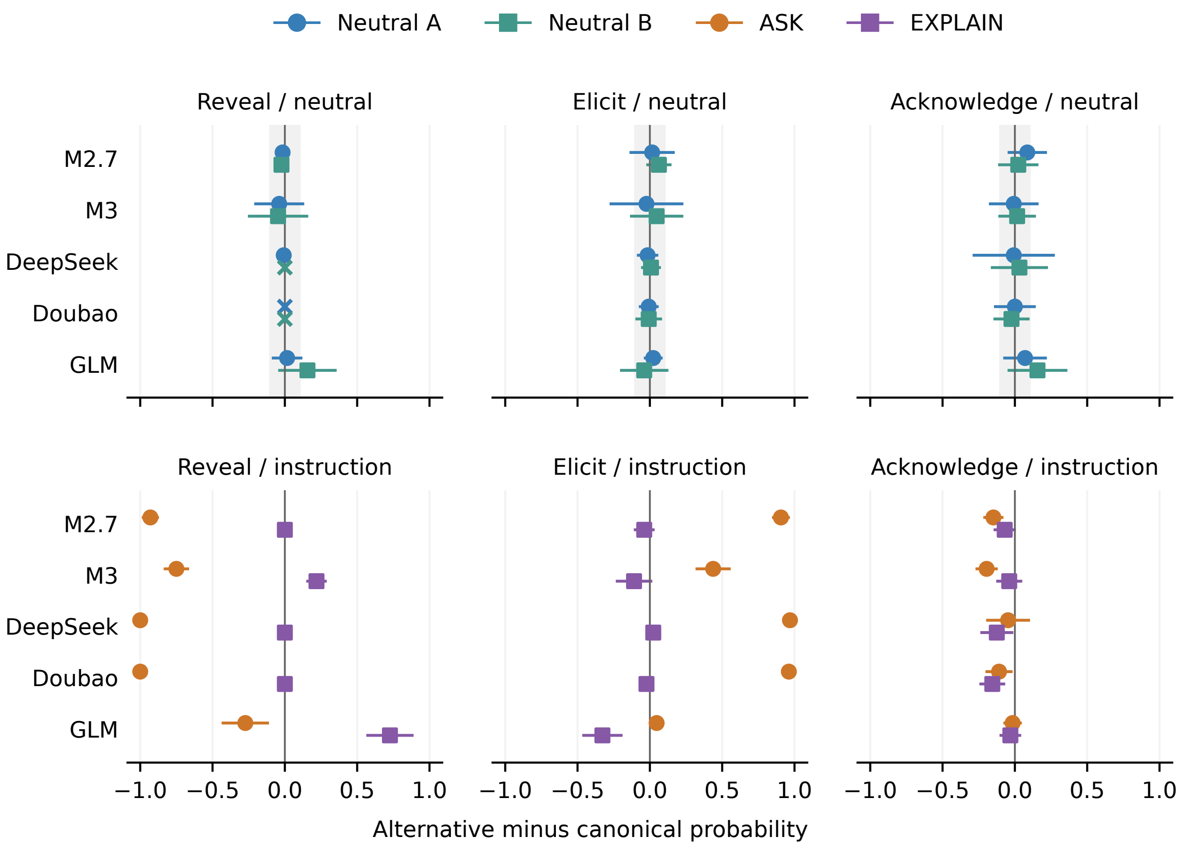

Figure 8: Paired changes relative to canonical wording on the same 16 sources. Neutral arms show nominal Bonferroni simultaneous 95% paired-t intervals over 30 contrasts, with a ± 0.10 reference band. Explicit instruction arms show descriptive 95% paired-t intervals. Crosses identify zero-variance intervals; sparse-outcome undercoverage is not limited to those cases. These intervals cannot establish population equivalence; the bounded-source check below controls that interpretation.

Deployment	Reveal	Elicit	Acknowledge
MiniMax-M2.7	99.6	8.2	27.0
MiniMax-M3	83.2	48.4	69.9
DeepSeek-V4-Pro	100.0	2.0	59.0
Doubao-Seed-2.0-Lite	100.0	4.3	60.9
GLM-5.3	22.3	96.5	51.2

Table 10: Canonical event rates (%): 256 answers per deployment, eight per source, on 32 sources. Events can co-occur. Abbreviated deployment names in the text refer to these configurations, not all releases in their model families.

Predictor	Reveal	Elicit	Acknowledge
Context	0.153592	0.218731	0.182418
Context + model default	0.062317	0.089033	0.162597
+ model $\times$ subject	0.061894	0.092552	0.160458
+ model $\times$ student state	0.061463	0.092714	0.150724
Context + training style	0.138611	0.178259	0.178639

Table 11: Absolute source-averaged Brier losses on the 1,280 canonical confirmation responses. Lower is better. The style competitor is descriptive.

M2.7 2/17, M3 3/20, DeepSeek 0/0, Doubao 0/2, and GLM 0/10, each out of 256 responses. These are reviewer flags, not an accuracy ranking. Overall reviewer verdict agreement is 96.6%, but error-positive agreement is only $2(5)/(49+5) = 18.5\%$, illustrating the effect of rare positives. Each reviewer matches all six error and six no-error agent-authored controls; these limited diagnostics do not resolve natural-response disagreement.

Across the three screens, the default-gain intervals remain above zero for all three actions. Conditional acknowledgement intervals are [0.00551, 0.01405] under unanimous-error exclusion and [0.00510, 0.01472] under either-error exclusion. Conditional revelation and elicitation intervals span zero under every screen. The last two screens are identical and are not independent checks. Screening conditions on output judgments, changes the retained target, and retains source-equal weighting; it does not remove ability as a cause. The full primary tests are unchanged. The content-independent checkpoint's all-canonical and mathematics tables agree with their completed counterparts within 10^{-12}, using the same routines rather than an independent implementation.

F.6 Interpretation of State Profiles and Sensitivities

The acknowledgement profiles clarify the positive conditional result. For correct unfinished work, Doubao acknowledges difficulty in 0/64 calm responses and 60/64 frustrated responses; DeepSeek changes from 22/64 to 34/64. For erroneous work, their corresponding changes are 32/64 to 64/64 and 46/64 to 49/64. These are responses to scripted text cues, not measurements of empathy or learner understanding. The descriptive curves appear in Appendix F; the locked prediction comparison provides the test that such model-specific patterns recur on new sources.

Eight coding panels preserve the broad separation between default and student-state increments. Default-gain estimates range from 0.08571–0.09562 for revelation, 0.12442–0.13111 for elicitation, and 0.01699–0.02017 for acknowledgement. Conditional acknowledgement gains range from 0.00697–0.01052; conditional revelation and elicitation remain near zero. These reuse the same outputs and are sensitivity estimates, not independent replications.

Effect strength depends materially on which deployments are included. Deleting GLM reduces default gains to 0.00394 for revelation and 0.03080 for elicitation. Deleting M2.7 reduces the acknowledgement default gain to 0.00175, with an interval crossing zero [-0.00174, 0.00547]. The full-panel findings therefore do not imply equally strong differences among every subset of models. Across 200 training-source resamples, the descriptive 2.5–97.5 percentile range for conditional acknowledgement gain is [0.00542, 0.01039]; revelation spans zero and elicitation remains slightly negative. These distributions condition on fixed confirmation labels and the original regularization ratio.

The six confirmation budget exceptions do not

Event	Increment	Brier gain	95% bootstrap interval	Holm p
Reveal	Default	0.091275	[0.071365, 0.109169]	5.12×10^{-10}
Reveal	Student state	0.000431	[-0.000373, 0.001299]	0.325749
Elicit	Default	0.129698	[0.121308, 0.137834]	3.88×10^{-24}
Elicit	Student state	-0.000163	[-0.001007, 0.000726]	0.640847
Acknowledge	Default	0.019821	[0.014020, 0.025270]	3.00×10^{-7}
Acknowledge	Student state	0.009734	[0.005363, 0.013947]	0.000192

Table 12: Six primary comparisons on 32 sources. Default adds model identity to context; student state adds model-specific student-state responses to the model-by-subject predictor. Tests are one-sided paired source tests with Holm correction across six; bootstrap intervals are individual 95% intervals.

Deployment	Event	Canonical	Neutral A	Neutral B	ASK	EXPLAIN
M2.7	Reveal	100.0	98.4	97.7	7.0	100.0
	Elicit	9.4	10.9	15.6	100.0	5.5
	Acknowledge	25.0	33.6	27.3	10.2	18.0
M3	Reveal	78.1	74.2	73.4	3.1	100.0
	Elicit	56.2	53.9	60.9	100.0	45.3
	Acknowledge	68.0	67.2	69.5	48.4	64.1
DeepSeek	Reveal	100.0	99.2	100.0	0.0	100.0
	Elicit	3.1	1.6	3.9	100.0	5.5
	Acknowledge	50.8	50.0	53.9	46.1	38.3
Doubao	Reveal	100.0	100.0	100.0	0.0	100.0
	Elicit	3.9	3.1	3.1	100.0	1.6
	Acknowledge	58.6	58.6	56.2	47.7	43.0
GLM	Reveal	27.3	28.9	43.0	0.0	100.0
	Elicit	95.3	97.7	91.4	100.0	62.5
	Acknowledge	49.2	56.2	64.8	47.7	46.1

Table 13: Matched prompt-arm event rates (%). Every cell has 128 answers on the same 16 sources. Abbreviations refer to the five specified deployments.

Deployment	Explain	Choice	Qualify	Request	Ability claim
M2.7	98.0	11.7	4.7	0.0	6.6
M3	92.2	25.8	5.9	0.0	18.0
DeepSeek	96.9	4.3	9.4	0.0	2.0
Doubao	99.2	3.9	3.1	0.0	1.2
GLM	78.9	3.9	4.3	0.0	2.3

Table 14: All five secondary canonical event rates (%), each based on 256 answers. Columns denote worked explanation, learner choice, epistemic qualification, information request, and unsupported ability attribution. Measurement quality varies by event; these are not validated personality axes.

Subset	Sources/answers	D: R	D: E	D: A	S: R	S: E	S: A
All canonical	32/1280	0.09127	0.12970	0.01982	0.00043	-0.00016	0.00973
Mathematics	8/320	0.11624	0.13892	0.01992	0.00062	0.00030	0.01380
Exclude unanimous	32/1255	0.09234	0.12948	0.02010	0.00047	-0.00017	0.00987
Exclude either	32/1065	0.09535	0.13179	0.02068	0.00076	0.00023	0.00989
Exclude either/uncertain	32/1065	0.09535	0.13179	0.02068	0.00076	0.00023	0.00989

Table 15: All five content-sensitivity subsets, using unchanged locked predictions. D denotes default versus context; S denotes student state versus model-by-subject. R/E/A denote revelation, elicitation and acknowledgement. Entries are source-equal Brier gains, not new hypothesis tests.

determine the primary findings. For either binary value of each affected canonical GLM code, all six primary estimates and adjusted p-values are exactly unchanged: the other two coders agree on the three primary events. This fixed-prediction enumeration addresses those missing codes only. It does not validate coder behavior across the Gateway interruption. Complete secondary results and sensitivity ranges are retained in Appendix F.

Mathematics and content screening. The mathematics subset (eight sources, 320 responses) retains default gains of 0.11624, 0.13892, and 0.01992; student-state increments are 0.00062, 0.00030, and 0.01380 for revelation, elicitation, and acknowledgement. All 2,584 content reviews are valid after the disclosed structural repairs. Excluding five-model slots flagged by either reviewer retains 1,065 responses on all 32 sources. Default gains are 0.09535, 0.13179, and 0.02068; conditional acknowledgement gains 0.00989 [0.00510, 0.01472], while the other two conditional intervals span zero. The unanimous-error screen retains 1,255 responses and the same qualitative pattern. No uncertainty verdicts occur, so the error-or-uncertainty screen is identical to the either-error screen (Appendix F.5).

These are descriptive selected-subset estimates, with no new tests. Both reviewers match all twelve diagnostic controls, yet agree on an error in only 5 of the 49 natural responses flagged by either (positive agreement 18.5%). Screening thus tests sensitivity to these judgments; it neither guarantees correct remaining answers nor causally separates behavior from ability.

G Expression-Transfer Extension and Measurement History

The extension protocol and forecast table were frozen on 7 September 2026 before its first tutor generation. Its 32 problem sources had already been observed in the first confirmation stage; they are not a second independent source holdout. Each of four expression arms crosses 32 sources, two student-work states, two cue poles, five deployments, and two requests, yielding 1,280 answers. Explicit and implicit arms each assign one of four fixed phrase pairs per source, balanced with two sources per domain and phrase pair. Student-work text and reference targets are inherited. The peer arm assigns the current student the calm state in the frozen predictor regardless of the peer's quote. Exact materials, the 61,440-row forecast table covering all endpoints and predictors, and their hashes are retained with the executable analysis.

The extension yields all 5,120 answers and uses the original three event coders and evidence-line rubric. After the recovery described below, all 15,360 codes are usable. Pairwise positive coder agreement is 99.25–99.41% for revelation, 90.68–92.34% for elicitation, and 94.89–95.51% for acknowledgement. Of 15,360 requested codes, 15,359 pass the original strict parser after the fixed repair budget. The remaining final response contains all eight complete event objects but lacks one root closing brace. A separately recorded amendment appends exactly that brace to a derived copy, preserving raw responses, earlier failed attempts, and the failed controller's exit record. No event label or evidence is edited and no additional API request is made. The other two coders agree on all three primary events for this response; replacing the missing vote by either binary value leaves every primary majority unchanged. Two secondary events do not have this invariance. Derived measurement selection, original diagnostics, and recovered diagnostics are stored separately.

Generation completed on 7 September and measurement recovery on 8 September 2026. The first-stage fitted predictors remain unchanged. Six new-expression primary tests form their own Holm family; contemporary original-wording comparisons, expression-minus-original paired contrasts, phrase subgroups, peer contrasts, repetition, and coding/deletion sensitivities are secondary. Bootstrap intervals use 10,000 source resamples stratified by domain with seed 20260907. They condition on the fixed training fit and the finite source/phrase assignment. The individual-coder panels change measurements; generator-deletion panels remove scored observations without refitting the original predictors.

Contemporary original-wording control. The acknowledgement state gain is +0.01248 with a 95% source-bootstrap interval of [0.00888, 0.01624] when the original wording is generated again. Paired new-minus-contemporary gain differences are -0.01614 [-0.02254, -0.00990] for explicit synonyms and -0.01876 [-0.02384, -0.01358] for implicit cues. These secondary contrasts compare expression conditions during the same observation period; they do not isolate the effects of provider changes over time.

Expression	Forecast increment	Gain [95% CI]	Holm p
Explicit	Default: reveal	.09730 [.08095, .11320]	7.63×10^{-11}
Explicit	Default: elicit	.13070 [.12266, .13825]	4.97×10^{-24}
Explicit	State: acknowledge	-.00366 [-.00793, .00090]	1.0
Implicit	Default: reveal	.08231 [.06657, .09695]	2.76×10^{-9}
Implicit	Default: elicit	.12748 [.11660, .13806]	8.51×10^{-20}
Implicit	State: acknowledge	-.00628 [-.00960, -.00273]	1.0

Table 16: Six frozen-forecast primary expression-transfer tests. Gains are source-averaged reductions in Brier loss; intervals bootstrap 32 reused source problems, conditioning on fitted forecasts. The default baseline is educational context; the state baseline is the model-by-domain profile. One-sided tests ask whether the gain is positive: $p = 1$ is not a separately specified test of harm, and the confidence intervals are not simultaneous intervals.

Arm / pair	Calm pole	Frustrated pole
Original	I feel calm while working on this.	I feel frustrated while working on this.
Explicit 1	I'm feeling at ease as I work through this.	I'm feeling annoyed as I work through this.
Explicit 2	Working on this leaves me feeling relaxed.	Working on this leaves me feeling irritated.
Explicit 3	I'm feeling composed about working on this problem.	I'm feeling exasperated about working on this problem.
Explicit 4	I feel settled while I work on this.	I feel fed up while I work on this.
Implicit 1	I've set aside some time for this; one step at a time is fine.	Ugh, this again. I'm about ready to close the page.
Implicit 2	I'm taking my time with the page. There's no rush.	I've just pushed the page away. Ugh, enough already.
Implicit 3	I'm sitting back with my notebook open, ready to take the next step.	I've thrown my pencil down beside the notebook. Ugh.
Implicit 4	I'll take a slow breath and carry on with the page in front of me.	I'm gritting my teeth at the page in front of me. Ugh.
Peer	I feel calm while working on this. My classmate, who is doing a different problem, says, "I feel calm while working on this."	I feel calm while working on this. My classmate, who is doing a different problem, says, "I feel frustrated while working on this."

Table 17: Complete fixed expression stimuli. Pole labels denote the intended stimulus contrast, not diagnosed human emotion. Within each domain, sources are ordered by their frozen hash: explicit pair index cycles through 1–4; implicit pair index is offset by one modulo four. Each source keeps its assigned pair across student-work states, models, and requests.

Peer-context control. Keeping the current student explicitly calm, changing a peer's quote yields acknowledgement-rate changes of +30.47, +25.00, +26.56, +23.44, and -4.69 percentage points for M2.7, M3, DeepSeek, GLM, and Doubao, respectively. The first four source intervals exclude zero; Doubao's includes it. These are secondary descriptive contrasts. One deterministically selected 0-to-1 example per model helps interpret the event: the first four models explicitly affirm the current student's calmness, sometimes correctly distinguishing the peer's frustration. The event-rate contrast therefore cannot itself establish misattribution. Outcome-selected examples do not estimate its prevalence.

H Recovered Archive Inputs and Provenance Limits

A local reconstruction of six tutor-dialogue settings recovers 3,454 task items and 960 exact question/conversation source groups, including 940 groups shared across benchmark settings. Of these, 551 items in 250 groups lack a separately recovered target question; the remaining 710 groups have at least one known problem. Reference fields comprise 2,679 provided non-placeholder solutions, 475 placeholders, and 300 absent references. Provided solutions are not thereby verified as correct. Exact grouping normalizes Unicode and whitespace, not mathematical content or paraphrase equivalence, and does not certify statistical independence.

All 92 prediction-file and summary pairs match the earlier census snapshots. The reconstruction links 54,264 latest-success file-item records without unmatched joins, yielding 46,356 distinct variant-preserving response-text records. A seven-model standard-variant panel contains 24,178 records. Reviewed historical adapter methods support reconstruction consistency, but do not attest all historic data, constants, client behavior, or deployed weights.

A separate protocol selects 256 eligible exact source groups with paired scaffolding/pedagogy replies from seven historical model aliases: 3,584 existing answers, requiring 10,752 automated codes. Source-grouped cross-fitting and paired instruction effects were specified before coding. Coding on 8 September 2026 produced 10,750 valid codes; two provider-blocked codes remain absent. The complete-coding gate failed, but the two available votes identify every affected majority regardless of the missing vote. Appendix K reports the resulting analysis and its post-hoc missing-data amendment.

An input-only scope audit reproduces the frozen selection exactly. The standard scaffolding pool has 1,150 items in 741 exact source groups; excluding 148 items with missing problems and placeholder references leaves 1,002 items in 612 groups. Source-level selection chooses 256 groups and one dialogue per group. The selected contexts contain 243 two-utterance, twelve three-utterance, and one four-utterance histories. Median context lengths are 398 characters in the full pool, 422 in the eligible pool, and 429 in the selection. These summaries do not establish representativeness of classroom interactions or the full archive.

MathTutorBench combines Bridge's real tutoring snippets (Wang et al., 2024) and MathDial's teacher–simulated-student dialogues (Macina et al., 2023) into its MathDialBridge collection (Macina et al., 2025). All selected problem strings exactly match questions in the local MathDial inventory, but bridge records lack per-item upstream dataset identifiers. Question overlap does not attest exact dialogue lineage or actual learner participation. Accordingly, we refer to archived benchmark dialogue rather than real-classroom validation. Existing inputs and references may be human-authored; upstream human quality/preference labels and the omitted target tutor utterance are not used as behavioral ground truth.

I Post-Result Diagnosis of the Transfer Contrast

This diagnostic was specified after the extension's primary results and absolute-loss summaries had been inspected. It adds no API calls, fitted parameters, hypothesis tests, or independent confirmation. The three student expression arms share identical frozen predictions at every matched source, domain, model, student-work state, cue pole, and repetition index. Peer controls are excluded because their own-student affect assignments differ.

For baseline probability b, conditional probability e, contemporary label y_0, and new-expression label y_1, let $d = e - b$ and $z = y_1 - y_0$. The change in Brier-loss gain has the finite-panel identity

$$\Delta G = 2\,\mathrm{E}[zd] = 2\,\mathrm{E}[z]\,\mathrm{E}[d] + 2\,\mathrm{Cov}(z, d). \quad (1)$$

Expectations and covariance use equally weighted observed cells, not a causal model. For acknowledgement, mean conditional-minus-domain probability is 0.000009812. The overall-rate terms are approximately -0.000001043 for explicit synonyms and -0.000002162 for implicit cues, compared with total gain changes of -0.016144 and -0.018761. A common prevalence shift alone therefore cannot account for this relative contrast. This does not evaluate the generalization of any recalibrated predictor or exclude model/cue-specific recalibration.

M3 and DeepSeek show larger, not absent, acknowledgement cue gaps under the new expressions. Their weighted contributions to the total explicit gain change are -0.005577 and -0.010557, and to the implicit change -0.005382 and -0.013470. M2.7 and GLM contribute in the positive direction, and Doubao in the negative direction. Full model, domain, work-state, and cue-pole contributions are retained and sum to the respective total within numerical precision. These are descriptions of the same outputs, not causal effects or independent replications. New and contemporary replies need not share random seeds for the accounting identity; their pairing does not represent a real learner's trajectory. The two expression arms' different absolute-loss changes and the finite cue bank continue to limit a single-mechanism interpretation.

J Source-Held-Out Calibration Diagnosis

After inspecting the primary results and absolute-loss diagnostic, we specify a separate acknowledgement-only check. The domain and conditional predictors receive identical calibration procedures. The identity map retains original probabilities. Shared logistic calibration fits $\sigma(a + b\,\mathrm{logit}(p))$ across models; model-specific calibration fits one such map per deployment. We clip input probabilities to $[10^{-6}, 1 - 10^{-6}]$ for logistic maps, minimize the summed binomial

Deployment	Frozen cue gap	Original	Explicit	Implicit
MiniMax-M2.7	.5417	.3516	.4375	.4531
MiniMax-M3	.3597	.3281	.7656	.7500
DeepSeek-V4-Pro	.2625	.1875	.4844	.5078
Doubao-Seed-2.0-Lite	.7455	.7500	.7422	.6484
GLM-5.3	.6788	.6094	.6406	.8438

Table 18: Acknowledgement-rate gaps: frustrated pole minus calm pole, averaged across matched problems, work states, and requests. The frozen conditional probability gap is identical across expression arms; the other columns are observed gaps. These are post-result descriptive contrasts without new tests.

negative log likelihood plus $\frac{1}{2}[a^2 + (b-1)^2]$, and fix $a \in [-10, 10]$, $b \in [0, 5]$ without tuning. All 240 fitted maps converge without reaching these bounds.

Each domain's eight source families are hash-ordered and assigned cyclically to four folds. Every model, state, repetition, and expression for a source shares its fold. Each fit uses 24 sources and predicts eight held-out sources. The original-wording training regime only uses contemporary original-wording labels; the target-arm regime uses labels from the target expression. The latter evaluates adaptation to source-held-out inputs, not zero-shot expression transfer. Source separation does not guarantee phrase separation in this nested material bank. Tests verify that held-out source labels cannot change their predictions and target-arm labels cannot affect original-wording calibration.

Under target-arm per-model calibration, explicit conditional loss improves from 0.144501 to 0.132657, while domain loss improves from 0.140840 to 0.129123. Implicit conditional loss improves from 0.155411 to 0.133293, while domain loss improves from 0.149134 to 0.125062. Adaptation can therefore improve absolute prediction without restoring the old interaction's incremental advantage. Per-model calibration can itself absorb model-dependent cue responses, so successful baseline adaptation is not evidence of their absence.

All absolute losses, parameters, source predictions, and domain-stratified source bootstrap intervals are retained. The 10,000 draws use seed 20260908; intervals condition on cross-fitted forecasts without refitting each bootstrap or accounting for the full post-result selection history. There are no new hypothesis tests. These two calibration families constrain a specific alternative explanation; they do not exclude nonlinear maps, other features, training objectives, or larger adaptation samples.

K Archived Dialogue: Identified Majorities and Instruction Transfer

Sample and analysis chronology. The input-only audit above defines 256 exact source groups, one dialogue per group, paired across MathTutorBench standard scaffolding and pedagogy tasks (Macina et al., 2025). Both instructions require at most two sentences. Scaffolding asks for useful, caring mathematics tutoring; pedagogy explicitly requests stepwise guidance, probing, and one question at a time. Seven historical deployments yield 3,584 archived replies. GLM-5.2 and Doubao-2.0-Pro are distinct from the GLM-5.3 and Doubao-Lite configurations in the prospective study. Historical model aliases are not evidence of immutable weights. No original tutor gold utterance or human teaching-quality label is used as a behavioral target.

The selection, three core events, grouped prediction, and paired instruction contrasts were fixed before new coding, after the prospective study's results were known. This is retrospective analysis of existing output, not prediction locked before those tutor replies were generated. Current MiniMax-M3, DeepSeek-V4-Pro, and GLM-5.3 code the visible answers using the same event rubric.

Two absent codes and exact majority identification. Three fixed passes produced 10,750 structurally valid codes of 10,752 requested. Two DeepSeek requests retained HTTP 400 errors. A disclosed post-limit diagnostic allowed exactly one identical-payload attempt per missing request, preserving all valid codes. Both returned the provider code `SensitiveContentDetected`. No input was rewritten, no alternative route was used, and no further attempts were made. The original complete-coding gate therefore failed.

For each of the two affected replies, the other two coders agree on all eight events. With n observed votes and s positives, a three-vote majority lies in $[\mathbf{1}(s \geq 2), \mathbf{1}(s + 3 - n \geq 2)]$. These

Training arm	Map	Original	Explicit	Implicit
None	Identity	.012483	-.003661	-.006277
Original	Shared logistic	.012130	-.002864	-.005871
Original	Per-model logistic	.005646	-.000846	-.003864
Target	Shared logistic	—	-.005239	-.010456
Target	Per-model logistic	—	-.003534	-.008231

Table 19: Post-result acknowledgement Brier gains: calibrated domain loss minus calibrated conditional loss, with the same map class applied to both. Original means contemporary original wording. Target-arm calibration uses target labels from other sources. Negative estimates do not all have intervals excluding zero.

endpoints coincide for all sixteen affected events. All 28,672 response–event majorities are thus identified without imputing the missing coder labels. Three-coder unanimity remains unknown for those sixteen events. This amendment was specified after inspecting the two available votes; it is not a preregistered missing-data rule. The original failed runs, incomplete coverage, and separate identified-majority analysis are retained.

Prediction and uncertainty. Five folds are assigned by sorting source hashes under a fixed salt and taking index modulo five. Every model, instruction, and event from a source shares its fold. Training rates use $(s + 0.5)/(n + 1)$ without tuning. The instruction-only baseline pools models in the target instruction; the same-instruction model rate uses that model and instruction. The other-instruction rate uses the same model under the other instruction, excluding the entire target source fold. The latter predictor does not use target-instruction labels, but its comparison baseline does. This is not a wholly target-instruction-unseen evaluation.

Losses average equally over models within each source, then over 256 sources. Table 21 reports all twelve pre-specified comparisons. Ten thousand source bootstrap draws use seed 20260908 and fixed cross-fit probabilities. These descriptive intervals omit refitting uncertainty and do not fully capture dependence induced by shared training folds; no additional confirmatory p-values are claimed. Negative other-instruction gains can reflect a shift in overall action rates, without establishing a reversal of model rankings.

Absolute errors, coding, and sensitivity. For scaffolding, baseline/model Brier losses are 0.24025/0.19344 for revelation and 0.23462/0.19269 for elicitation; pedagogy gives 0.06504/0.06423 and 0.05577/0.05537. Complete three-coder replies number 3,582. Positive agreement is 96.53–98.29% for revelation, 97.61–98.06% for elicitation, but only 65.55–66.67% for acknowledgement. The latter occurs in 27/1,792 scaffolding and 25/1,792 pedagogy replies. High overall agreement on this rare event does not establish accurate positive detection or absence of affect responsiveness.

Excluding the two affected source groups removes all their models and both instructions, leaving 254 complete sources. Each coder and the majority panel are separately refitted using the same five-fold hash rule, now applied to 254 sources. Scaffolding revelation gains across individual coders are 0.04429–0.04645 and elicitation gains 0.03980–0.04068; majority gains are 0.04650 and 0.04116. Pedagogy gains remain small and revelation/elicitation cross-instruction gains remain negative. These are post-hoc complete-source sensitivities. On the full majority panel, excluding each generator while retaining the original forecasts yields scaffolding gains of 0.03055–0.05485 and 0.02716–0.04910; all corresponding conditional interval lower bounds are positive. This fixed-forecast exclusion is not a smaller-panel refit.

All 21 paired instruction differences, joint action rates, absolute losses, source predictions, and sensitivity panels are retained in the accompanying analysis tables. The mostly two-utterance benchmark contexts and incompletely attested historical requests support a limited archived-dialogue extension, not classroom validation, a human personality taxonomy, or learning benefits.

Full-source individual-coder bounds. A final post-hoc check restores the full 256-source panel for individual-coder sensitivity without claiming to recover the absent labels. For DeepSeek, each primary event has two unknown votes; we refit forecasts under all four possible assignments. Since event predictions and Brier losses are separable, these ranges cover all 64 joint assignments of the six missing primary labels. MiniMax-M3 and GLM-5.3 have complete observed labels. Across

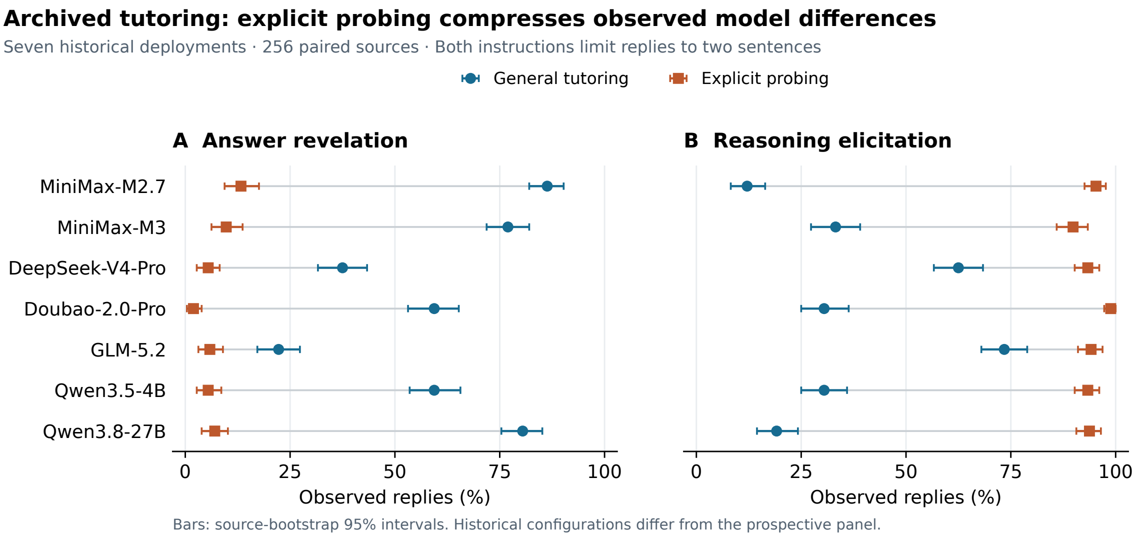

Figure 9: Paired archived action profiles on 256 sources. Explicit probing compresses the observed range across seven historical deployments. Lines connect the same model under two instructions; bars are the existing source-bootstrap 95% intervals for individual rates, not intervals for paired changes. Both instructions limit replies to two sentences. This historical panel differs from the prospective panel; all rates and prediction comparisons are reported below.

all coders and assignments, scaffolding revelation gains range from 0.04476 to 0.04667 and elicitation gains from 0.04046 to 0.04153; their conditional interval lower bounds remain positive. All cross-instruction revelation/elicitation gains remain negative. Pedagogy revelation gains are only 0.00074–0.00079; elicitation ranges from -0.00034 to 0.00054 across coders and assignments. The latter is therefore not a uniformly positive coder-level result. Hypothetical votes are stored separately as sensitivity scenarios, never as observed codes.

Historical deployment	Instruction	Reveal (%)	Elicit (%)	Acknowledge (%)
DeepSeek-V4-Pro	Scaffolding	37.50	62.50	3.91
DeepSeek-V4-Pro	Pedagogy	5.47	93.36	1.17
Doubao-2.0-Pro	Scaffolding	59.38	30.47	1.56
Doubao-2.0-Pro	Pedagogy	1.95	98.83	1.95
GLM-5.2	Scaffolding	22.27	73.44	1.56
GLM-5.2	Pedagogy	5.86	94.14	1.56
MiniMax-M2.7	Scaffolding	86.33	12.11	0.78
MiniMax-M2.7	Pedagogy	13.28	95.31	0.78
MiniMax-M3	Scaffolding	76.95	33.20	0.78
MiniMax-M3	Pedagogy	9.77	89.84	1.17
Qwen3.5-4B	Scaffolding	59.38	30.47	1.56
Qwen3.5-4B	Pedagogy	5.47	93.36	1.56
Qwen3.8-27B	Scaffolding	80.47	19.14	0.39
Qwen3.8-27B	Pedagogy	7.03	93.75	1.56

Table 20: Archived event rates: 256 replies per historical deployment and instruction. Events may co-occur. Pedagogy specifies probing; scaffolding gives a general useful/caring instruction. These aliases do not denote the same complete panel as in the prospective experiment.

Target instruction	Event	Model rate from	Brier gain	Lower	Upper
Pedagogy	Acknowledge	Same instruction	-0.00004	-0.00007	0.00001
Pedagogy	Acknowledge	Other instruction	-0.00012	-0.00022	0.00000
Pedagogy	Reveal	Same instruction	0.00081	0.00013	0.00156
Pedagogy	Reveal	Other instruction	-0.32304	-0.34579	-0.29884
Pedagogy	Elicit	Same instruction	0.00041	-0.00006	0.00092
Pedagogy	Elicit	Other instruction	-0.36563	-0.38669	-0.34182
Scaffolding	Acknowledge	Same instruction	0.00005	-0.00009	0.00021
Scaffolding	Acknowledge	Other instruction	-0.00003	-0.00007	0.00002
Scaffolding	Reveal	Same instruction	0.04681	0.03834	0.05528
Scaffolding	Reveal	Other instruction	-0.27443	-0.31452	-0.23383
Scaffolding	Elicit	Same instruction	0.04193	0.03385	0.05011
Scaffolding	Elicit	Other instruction	-0.32111	-0.36034	-0.28184

Table 21: All archived prediction comparisons. Positive gain means lower Brier loss than the target instruction-only baseline. Bounds are descriptive 95% source-bootstrap intervals conditional on the cross-fit predictions.